

Optimal Policy Design in Nonlinear DSGE Models: An n -order Accurate Approximation

By ISAAC GROSS* AND JAMES HANSEN†

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We derive an n -order accurate approximation of optimal policy for a wide class of nonlinear DSGE models analytically. Using Taylor polynomials to approximate welfare and the equilibrium, the $(n, n+1)$ approximation relaxes symmetry in the objective and certainty equivalence in the solution, as implied by a Linear-Quadratic (LQ) approximation with $n = 1$. When $n > 1$, we illustrate how curvature in preferences and the constraints can affect optimal policy, deriving a solution that is n -order accurate as opposed to first-order accurate only. Comparing solutions when $n = 2$ (a Quadratic-Cubic approximation) and $n = 1$ (LQ), in a New Keynesian economy with nominal frictions, we find significant differences in the optimal response to shocks; the joint distributions of wage inflation, the output gap and nominal interest rates; welfare and accuracy.

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* Author: Gross: Department of Economics, Monash University, 900 Dandenong Road, Caulfield East, VIC 3145, Australia, isaac.gross@monash.edu.

† Corresponding author: Hansen: Department of Economics, University of Melbourne, 111 Barry Street, Carlton VIC 3010, james.hansen@unimelb.edu.au.

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Optimal Policy Design in Nonlinear DSGE Models

Nonlinearity is pervasive in many optimal policy problems. Yet, fully understanding its implications for optimal policy design can be difficult. Models that admit closed-form solutions are often used to gain intuition, or its role is studied numerically.

We propose a tool that offers new insights. Using Taylor polynomials, it allows one to study how nonlinearities affect optimal policy design *analytically*. Our contribution is to show how Taylor polynomials can be used to approximate the objective and the constraints in a wide class of smooth optimal policy problems, and where the solution is equivalent to an n -order accurate approximation of the underlying first-order conditions (FOC) of the original nonlinear optimal policy problem.¹

Polynomials are well understood. They can be used to reformulate the objective function over the domain of interest to the economist, and to derive analytical results that are quite general. By representing the objectives and the constraints through them, one can gain a better understanding of the policymaker's objectives, how constraints and frictions affect the trade-offs a policymaker may face, and the optimal response to shocks.

The most common example is the Linear-Quadratic (LQ) approximation, a first-order accurate ($n = 1$) approximation of optimal policy where n is the order of accuracy. LQ has been widely used in the analysis of optimal monetary and fiscal policy.² In New Keynesian models, it is used to rewrite household welfare in consumption and labor supply in terms of deviations in inflation and the output gap from their respective targets. One can then study the role of constraints, such as the Phillips curve, and make precise statements about the shocks that present a meaningful trade-off for policy, and the class of instrument or targeting rules that fall within the optimal class.³

However, one can go well beyond $n = 1$, and there are several reasons that this yields new insights. With LQ, the policymaker's objective is quadratic and thus symmetric.⁴ LQ assumes away any role for asymmetry that can otherwise arise from curvature in household or firm decision problems, and the policymaker always weights expansions versus contractions in the economy equally. This arises from the linear approximation desired, and is not a general property of the underlying nonlinear solution.⁵

¹Throughout, our interest will be in perturbation-based Taylor approximations (hereafter approximations).

²Recent examples include Garga and Singh (2021) and Bodenstein and Zhao (2020). For a more complete overview, see Benigno and Woodford (2012); Woodford (2010).

³See, for example, Giannoni and Woodford (2017); Benigno and Woodford (2005).

⁴By symmetry, here we are referring to the the objective function being symmetric about the y-axis (i.e. an even function), as opposed to symmetric in any permutation of its arguments.

⁵Asymmetry in preferences is also observed empirically, see for example Surico (2007).

With a quadratic objective and linear constraints, LQ implies certainty equivalence holds. Optimal policy is invariant to the distribution of shocks. Volatility does not matter, nor skewness nor kurtosis, nor changes in any higher-order moments over time. However, these cases are of interest and there is increasing evidence to suggest that such moments are important for business cycles (Salgado, Guvenen and Bloom, 2019; Bloom et al., 2018; Fernández-Villaverde et al., 2011)

Accuracy is also a consideration. While LQ is highly tractable, it can be inaccurate. This applies when there is strong curvature in preferences or the constraints (or both), and the variance and persistence of shocks is high.⁶ In numerical applications, second- or higher-order perturbations of optimal policy are now frequently used.⁷

We derive a more general class of approximating optimal policy problems that can be designed to achieve an approximation of *any* desired integer order of accuracy, $n \geq 1$. This approximation, an $(n, n + 1)$ approximation, relaxes the restrictions of a symmetric objective function, a solution that is invariant to the distribution of shocks and allows one to study the consequences for optimal policy of a solution that is n -order accurate, as opposed to first-order accurate only.

The $(n, n + 1)$ approximation uses an $n + 1$ -order Taylor expansion of the policymaker's objective function, and then eliminates the linear terms in that expansion using a weighted sum of $n + 1$ -order Taylor expansions of the backward- and forward-looking constraints that characterize behavior in the economy's decentralized equilibrium. One can then maximize the resulting modified objective subject to an n -order expansion of the same constraints. The solution to this approximating problem is equivalent to an n -order accurate approximation of the solution to the original nonlinear policy problem.

The $(n, n + 1)$ approximation's validity follows from how the Lagrange multipliers affect the solution in the smooth class of constrained optimization problems that we consider.⁸ With either backward- or forward-looking constraints, the Lagrange multipliers in the nonlinear problem are only present in the zero- and first-order terms in the n -order approximation of its solution. Thus, only linear terms in the objective of the approximating problem require elimination (and not higher-order terms), making the approach highly tractable.

⁶For example, this is shown in consumption asset pricing models (Caldara et al., 2012) and in the neoclassical growth model (Aruoba, Fernández-Villaverde and Rubio-Ramírez, 2006).

⁷See, for example, Kim and Ruge-Murcia (2011) and Faia (2009).

⁸Specifically, where the objective and constraints are all $n + 1$ -times continuously differentiable and the Implicit Function Theorem holds (see Jin and Judd (2002) for further detail). There are important exceptions though, which also affect the use of standard perturbation methods. These include where constraints are occasionally binding, or where there are singularities in a perturbation-based solution.

The solution to the $(n, n + 1)$ approximation is valid for any bounded sequences of shocks, well-defined parameterization of the nonlinear optimal policy problem, and whether the steady state of the economy is efficient or not.⁹ Being a perturbation-based solution, it can be easily integrated with standard techniques for solving and estimating higher-order models (Andreasen, Fernández-Villaverde and Rubio-Ramírez, 2018; Fernández-Villaverde, Rubio-Ramírez and Schorfheide, 2016).

To illustrate how higher-order effects are important, we study the simplest generalization going from a $(1, 2)$ approximation (i.e. LQ) to the $(2, 3)$ approximation (i.e. a Quadratic-Cubic or QC approximation) in a New Keynesian economy with costly price and wage adjustment. This model is well understood and even here the order of approximation matters for optimal monetary policy design. For example, under the QC approximation, shock variances affect the optimal nominal interest rate path, the Divine Coincidence may not hold, and there are asymmetries in the objective function that depend on curvature in the utility function *and* the constraints, none of which are apparent under LQ.

To quantify these differences, we parameterize the New Keynesian (NK) model using quarterly US data (1964:II to 2006:II). Much (but not all) of the nonlinearity in the model comes from household risk aversion and the asymmetry in nominal wage adjustment costs. Without them, the NK model is closer to being linear and the differences between the optimal policy responses under the LQ and QC approximations are smaller.

However, when wage adjustment costs are asymmetric (it is more costly to reduce nominal wages than to increase them), and households are more consumption risk averse than a logarithmic (log) utility function would imply (as we infer from the data), there are substantive differences in shape of the objective functions and the constraints that lead to different optimal policy responses. Impulse response functions vary greatly in magnitude, and in some cases even switch sign, and there is left truncation in the joint distribution of wage inflation and the output gap under QC, as the policymaker views contractions as more costly than expansions. With LQ, there is no such truncation as the distributions are always symmetric, and contractions and expansions are equally weighted.

We then consider the implications for household welfare. Under LQ, the usual conclusion holds that the welfare costs of using a Taylor rule are small. Relative to the steady state, the welfare gain in using optimal policy as opposed to a Taylor rule is only 0.05% of annual consumption. With QC, the implied welfare gains are roughly six times larger at 0.32% of

⁹That is, parameterizations where the solution is non-singular and where second-order sufficiency conditions are also satisfied (see online Appendix A.A1).

annual consumption. Calculating Euler equation errors to assess accuracy, these are smaller on average under QC than under LQ across a range of parameterizations of the models.

The next section reviews related literature and Section II outlines our theoretical contribution. Section III specifies the New Keynesian economy where we apply the $(n, n + 1)$ approximation with $n = 1$ (LQ) and $n = 2$ (QC). Section IV compares these problems analytically and Section V compares them quantitatively. The final section concludes.

I. Related Literature

The use of optimal policy approximations through polynomials is well established in optimal control theory (see for example, Bryson and Ho (1975); Fleming (1971)) with early economic applications such as Magill (1977). More recently, Benigno and Woodford microfound the LQ approximation ($n = 1$) in economic policy problems and apply the results in a series of papers and the approach is now widely used.¹⁰ LQ yields many novel insights including in open economies,¹¹ for optimal taxation and fiscal policy,¹² and on the interaction between monetary and macroprudential policy.¹³

Perhaps most closely related to our work, Benigno and Rossi (2021) study asymmetry using a second-order approximation of the New Keynesian model with log utility and Calvo pricing, although their emphasis is quite different. Rather than developing the underlying theory of higher-order optimal policy approximations as we do here, they focus on how price dispersion is an important source of asymmetry in the model that results in an expansionary bias in optimal policy overall (even if preferences would otherwise imply a *contractionary* bias). When this is combined with more volatile shocks, it leads to flatter estimates for the slope of a linear Phillips curve, when the true Phillips curve is in fact nonlinear.

Our work is also related to the literature on perturbation methods that is widely used to solve and estimate nonlinear DSGE models. Examples include the use of second-order perturbation to solve them (Schmitt-Grohé and Uribe, 2004*b*; Lombardo and Sutherland, 2007), to approximate time varying risk (Benigno, Benigno and Nisticó, 2013), and higher-order methods (Andreasen, Fernández-Villaverde and Rubio-Ramírez, 2018; Jin and Judd, 2002). An advantage of our approach is that being an n -order accurate perturbation-based solution as well, it can be easily integrated with existing numerical and estimation methods.

¹⁰For recent examples, see Bodenstein and Zhao (2020) and Garga and Singh (2021). Examples of Benigno and Woodford's applications include Woodford (2010); Benigno and Woodford (2005); Rotemberg and Woodford (1997).

¹¹See for example Corsetti, Dedola and Leduc (2010); De Paoli (2009); Galí and Monacelli (2005).

¹²See for example Benigno and Woodford (2006, 2004).

¹³See Hansen (2018); Nisticó (2016); Cúrdia and Woodford (2016).

II. Approximating Optimal Policy

Consider the general class of time-consistent optimal policy problems:

$$\begin{aligned} \max_{\{y_t\}_{t \geq t_0}} E_{t_0} & \left[\sum_{t \geq t_0} \beta^{t-t_0} f(y_{t-1}, y_t, \varepsilon_t) \right] \\ \text{s.t. : } 0 &= g_j(y_{t-1}, y_t, \varepsilon_t) \text{ for } j = 1, \dots, J \text{ and } \forall t \geq t_0 \\ 0 &= E_t[h_l(y_t, y_{t+1}, \varepsilon_{t+1})] \text{ for } l = 1, \dots, L \text{ and } \forall t \geq t_0 - 1 \end{aligned}$$

where $f : U \rightarrow \mathbb{R}$ is a scalar-valued objective function defined on an open set U in $\mathbb{R}^m \times \mathbb{R}^m \times \mathbb{R}^p$ and where y_t is an $m \times 1$ vector of time- t control variables (chosen by the policymaker) and ε_t is a $p \times 1$ vector of exogenous stochastic shocks (taken as given), $g_1, \dots, g_J$ denotes a set of backward-looking constraints with $g_j : U \rightarrow \mathbb{R}$ for $j = 1, \dots, J$ and $h_1, \dots, h_L$ denotes a set of forward-looking constraints with $h_l : U \rightarrow \mathbb{R}$ for $l = 1, \dots, L$, with $J + L < m$, and given initial conditions $y_{t_0-1}, \varepsilon_{t_0}$. We assume f , g_j and h_l are at least $n + 1$ -times continuously differentiable (i.e. are of class C^{n+1}) and that f is concave in $[y'_{t-1}, y'_t]'$ on U .

Define the stacked $(2m + p) \times 1$ vector of lagged controls, time- t controls and exogenous shocks, $\mathbf{x}_t := [y'_{t-1}, y'_t, \varepsilon'_t]'$,¹⁴ the nonlinear optimal policy (hereafter Ramsey) problem can be represented more compactly as:

$$\begin{aligned} (1) \quad \max_{\{y_t\}_{t \geq t_0}} E_{t_0} & \left[\sum_{t \geq t_0} \beta^{t-t_0} f(\mathbf{x}_t) \right] \\ \text{s.t. : } 0 &= g_j(\mathbf{x}_t) \text{ for } j = 1, \dots, J \text{ and } \forall t \geq t_0 \\ 0 &= E_t[h_l(\mathbf{x}_{t+1})] \text{ for } l = 1, \dots, L \text{ and } \forall t \geq t_0 - 1 \end{aligned}$$

Our focus is on policies that are *optimal from a timeless perspective* (Benigno and Woodford, 2012), as implied by the fact that we assume the forward-looking constraint set also holds in period $t_0 - 1$. This approach is often used with LQ and is equivalent to numerical approximation of the optimal Ramsey policy around the deterministic steady state.¹⁵ However, it does abstract from the question of *how* time-consistency is achieved. For brevity, we do not consider that question here, but note that one could alternatively modify the constraints on the time t_0 decisions, as Benigno and Woodford do, to study how a policymaker commits to its optimal path.

¹⁴Our notation is sufficiently general for y_t to include endogenous variables, such as consumption, output, investment and inflation, y_{t-1} to include endogenous pre-determined variables such as the capital stock and lagged consumption, and for exogenous shocks, ε_t to be augmented to include lagged exogenous shocks as well.

¹⁵See for example Faia (2009) and Schmitt-Grohé and Uribe (2004a)

The solution to (1) can be derived using the Lagrangian:

$$\Lambda_{t_0}^{Ramsey} = E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} \left[f(\mathbf{x}_t) - \sum_{j=1}^J \lambda_{jt} g_j(\mathbf{x}_t) - \sum_{l=1}^L \psi_{lt} h_l(\mathbf{x}_{t+1}) \right]$$

and taking the i^{th} optimality condition (FOC) w.r.t y_{it} (the i^{th} element of y_t):¹⁶

$$(2) \quad 0 = \frac{\partial f_t}{\partial y_{it}} + \beta E_t \left[\frac{\partial f_{t+1}}{\partial y_{it}} \right] - \sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} - \beta \sum_{j=1}^J E_t \left[\lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} \right] \\ - \sum_{l=1}^L \psi_{lt} E_t \left[\frac{\partial h_{lt+1}}{\partial y_{it}} \right] - \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}}$$

which must hold for all $i = 1, \dots, m$ and $t \geq t_0$, and when combined with the set of forward- and backward-looking constraints, provides the non-linear FOC for optimal Ramsey policy.

A. Multi-index Notation

To establish the link between an n -order approximation of the Ramsey FOC, and the solution to a polynomial approximating problem (what we will subsequently refer to as the $(n, n+1)$ approximation), we use multi-index notation. This notation provides a succinct method for writing Taylor expansions of an arbitrary order. Here we provide the main definitions we use.

A multi-index $\alpha_{\mathbf{z}} = (\alpha_1, \dots, \alpha_Z)$ is a Z -tuple of non-negative integers with $|\alpha_{\mathbf{z}}| = \alpha_1 + \alpha_2 + \dots + \alpha_Z$, where α_i refers to the i^{th} element of the multi-index $\alpha_{\mathbf{z}}$.¹⁷ Denoting the i^{th} element of a $Z \times 1$ vector $\mathbf{z}_t$ as z_{it} , we define the multi-index power $\mathbf{z}_t^{\alpha_{\mathbf{z}}} := \prod_{i=1}^Z z_{it}^{\alpha_i}$ and the factorial $\alpha_{\mathbf{z}}! := \prod_{i=1}^Z \alpha_i!$. The multi-index power of the derivative is:

$$\mathbf{D}^{\alpha_{\mathbf{z}}} := \left(\frac{\partial}{\partial z_{1t}} \right)^{\alpha_1} \left(\frac{\partial}{\partial z_{2t}} \right)^{\alpha_2} \dots \left(\frac{\partial}{\partial z_{Zt}} \right)^{\alpha_Z} = \prod_{i=1}^Z \left(\frac{\partial}{\partial z_{it}} \right)^{\alpha_i}$$

With this notation, an n -order Taylor expansion of a generic function $F : \mathbb{R}^Z \rightarrow \mathbb{R}$ around the approximation point $\mathbf{z}_t = \mathbf{z}$ can be represented as $F(\mathbf{z}_t) = \sum_{|\alpha_{\mathbf{z}}|=0}^n \frac{\mathbf{D}^{\alpha_{\mathbf{z}}}(F)|_{\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!}$ where $\mathbf{D}^{\alpha_{\mathbf{z}}}(F)|_{\mathbf{z}}$ denotes that the multi-index power of the derivative is taken over the function F and then evaluated at the point $\mathbf{z}_t = \mathbf{z}$ for a given multi-index $\alpha_{\mathbf{z}}$, and where $\tilde{\mathbf{z}}_t := \mathbf{z}_t - \mathbf{z}$. The sum $\left(\sum_{|\alpha_{\mathbf{z}}|=0}^n \right)$ is then taken over *all* multi-indices $\alpha_{\mathbf{z}}$ from $|\alpha_{\mathbf{z}}| = 0$ to $|\alpha_{\mathbf{z}}| = n$.

B. An n -order Accurate Approximation of the Ramsey FOC

Our interest is in a polynomial approximating problem that yields a solution that is equivalent to the n -order Taylor approximation of the Ramsey FOC (2), which are given by:

¹⁶Note we use the notation $\frac{\partial f_t}{\partial y_{it}} := \frac{\partial f(\mathbf{x}_t)}{\partial y_{it}}$ and $\frac{\partial f_{t+1}}{\partial y_{it}} := \frac{\partial f(\mathbf{x}_{t+1})}{\partial y_{it}}$.

¹⁷Throughout, we use a bold subscript to denote a particular class of multi-index, and non-bold subscripts when referring to an element of that multi-index.

(3)

$$0 = \sum_{|\alpha_{\mathbf{z}}|=0}^n \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\frac{\partial f}{\partial y_{it}} - \sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} - \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right) |_{\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \\ + E_t \left[\sum_{|\alpha_{\mathbf{z}}|=0}^n \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \frac{\partial f_{t+1}}{\partial y_{it}} - \beta \sum_{j=1}^J \lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} - \sum_{l=1}^L \psi_{lt} \frac{\partial h_{lt+1}}{\partial y_{it}} \right) |_{\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] + \mathcal{O}(\epsilon^{n+1})$$

for $i = 1, \dots, m$ and $t \geq t_0$, and the n -order Taylor approximation (hereafter approximation) of the backward and forward-looking constraints:

$$(4) \quad 0 = \sum_{|\alpha_{\mathbf{z}}|=0}^n \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} (g_{jt}) |_{\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} + \mathcal{O}(\epsilon^{n+1}) \text{ for } j = 1, \dots, J \text{ and } t \geq t_0$$

$$(5) \quad 0 = E_t \left[\sum_{|\alpha_{\mathbf{z}}|=0}^n \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} (h_{lt+1}) |_{\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] + \mathcal{O}(\epsilon^{n+1}) \text{ for } l = 1, \dots, L \text{ and } t \geq t_0 - 1$$

where the sum in each case is taken across all multi-indices $\alpha_{\mathbf{z}}$ such that $Z := 2m + p + 2J + 2L$, and from $|\alpha_{\mathbf{z}}| = 0$ to $|\alpha_{\mathbf{z}}| = n$, and where the $Z \times 1$ vector $\mathbf{z}_t = [y'_{t-1}, y'_t, \varepsilon'_t, \lambda'_{t-1}, \lambda'_t, \psi'_{t-1}, \psi'_t]'$ includes all lagged and current control variables $[y'_{t-1}, y'_t]'$, the shocks ε_t , and backward-looking $\lambda_t := [\lambda_{1t}, \dots, \lambda_{Jt}]'$ and forward-looking Lagrange multipliers $\psi_t := [\psi_{1t}, \dots, \psi_{Jt}]'$, $\tilde{\mathbf{z}}_t := \mathbf{z}_t - \mathbf{z}$, and $\mathbf{z}$ denotes the deterministic steady state value of $\mathbf{z}_t$. The remainder of the n -order approximation is denoted by $\mathcal{O}(\epsilon^{n+1})$, and the approximation is taken around the deterministic steady state and is (at least) valid locally in the neighborhood of $\mathbf{z}$ given bounded stochastic shocks, $\{\varepsilon_t\}_{t \geq 0}$.¹⁸

C. The $(n, n+1)$ Approximation

We now derive a polynomial approximation that has the aforementioned properties:

$$(6) \quad \max_{\{y_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} W_t^{(n, n+1)} \right] \\ s.t.: 0 = \sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt}) |_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+1}) \text{ for } j = 1, \dots, J \text{ and } \forall t \geq t_0 \\ 0 = E_t \left[\sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt+1}) |_{\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + \mathcal{O}(\epsilon^{n+1}) \text{ for } l = 1, \dots, L \text{ and } \forall t \geq t_0 - 1$$

¹⁸More precisely, following Benigno and Woodford (2012), we consider shock processes that satisfy $\varepsilon_t = \epsilon v_t$ where v_t is a bounded vector stochastic process and ϵ an arbitrarily small real number. The focus is on approximations that for any small enough ϵ , the equilibrium solution satisfies $\mathbf{z}_t(\epsilon) = \mathbf{z} + \mathcal{O}(\epsilon^n)$ where n is the approximation order and the approximation residual, $\mathcal{O}(\epsilon^n)$, becomes small in the sup norm (i.e. the least upper bound such that the norm of the residual vector, under the l^2 -norm for finite-dimensional vectors) is within that bound almost surely for all $t \geq t_0$.

and where the periodic $(n, n + 1)$ objective is given by:

$$(7) \quad W_t^{(n, n+1)} := - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{l=1}^L \psi_l \beta^{-1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(h_{lt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \\ + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + t.i.p + \mathcal{O}(\epsilon^{n+2})$$

and where the multi-index $\alpha_{\mathbf{x}}$ is a $2m + p$ tuple of non-negative integers and recall $\mathbf{x}_t := [y'_{t-1}, y'_t, \varepsilon'_t]'$ is a $(2m + p) \times 1$ vector of lagged controls (y_{t-1}) , time- t controls (y_t) and exogenous shocks (ε_t) , $\tilde{\mathbf{x}}_t := \mathbf{x}_t - \mathbf{x}$,¹⁹ and *t.i.p* denotes terms that are independent of policy. The non-standard terms in the $n + 1$ -order approximation of the objective are highlighted in blue as these provide useful intuition (discussed below).

We now state our main theoretical contribution:

PROPOSITION 1: *The FOC of the $(n, n + 1)$ approximating problem defined in (6) are equivalent to those derived from an n -order approximation of the FOC of the original Ramsey optimal policy problem defined by (3) to (5).*

The proof is detailed and we provide it (and all subsequent proofs) in the online Appendix. However, much of the intuition for it can be gleaned from the non-standard terms in the $(n, n + 1)$ approximation's objective, highlighted in blue, and that are used to eliminate the linear terms in $\tilde{\mathbf{x}}_t$. These terms are a weighted sum of the $n + 1$ -order expansions of the constraints, where each constraint is weighted by its respective steady state Lagrange multiplier obtained from the steady state solution to the nonlinear Ramsey FOC. Up to the approximating remainder, the terms in blue are equivalent to the *linear* terms in the $n + 1$ -order expansion of the objective (i.e. $\sum_{|\alpha_{\mathbf{x}}|=1}^1 \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} = \mathcal{T.I.B} + \mathcal{O}(\epsilon^{n+2})$ where $\mathcal{T.I.B}$ denotes the terms in blue in (7)).

When taking the derivatives of these highlighted terms with respect to the vector of controls (y_t) , one captures how the steady state values of the Lagrange multipliers, multiplied by the higher-order terms in each constraint, perturb the n -order approximation of the FOC of the corresponding *unconstrained* Ramsey policy problem. The higher-order effects of the Lagrange multipliers, which also perturb the approximated *unconstrained* Ramsey FOC, are then captured by the constraints in the $(n, n + 1)$ approximating problem.

The approximation is valid because in the nonlinear Ramsey FOC (i.e. (3) to (5)), the Lagrange multipliers only show up at the zero and first-order in any n -order approximation

¹⁹Note the distinction between $\tilde{\mathbf{x}}_t$ and $\tilde{\mathbf{z}}_t$ is the latter additionally includes the lagged and contemporaneous vectors of the Lagrange multipliers associated with the backward- and forward-looking constraints.

of them. Thus, these effects need only be accommodated to the first-order in the objective, and one *does not* need to eliminate any second- or higher-order terms when deriving the objective function in the $(n, n + 1)$ approximating problem. As discussed earlier, this result is well known for the LQ case (when $n = 1$), but it was unclear how to approach the problem for an arbitrarily accurate optimal policy approximation of integer order $n > 1$, as we consider here.

Turning to the economic implications of the $(n, n + 1)$ approximation, it should be clear the n -order approximation relaxes some of the restrictions implied by LQ. It is well known for example that LQ objective functions must be purely quadratic by design, and thus imply a form of symmetry where economic expansions and contractions will be equally weighted. However, this rules out asymmetries in preferences such as those found to be important for monetary policy (Surico, 2007), as well as asymmetry in the transmission of monetary policy that is commonly found empirically (Santoro et al., 2014).

Second, with LQ being a first-order accurate approximation, certainty equivalence holds. The variance or indeed any higher-order moments of shocks (or changes in these moments over time) have no effect on optimal policy. However, there is growing evidence to suggest both volatility and skewness are sources of business cycle fluctuations (Salgado, Guvenen and Bloom, 2019; Bloom et al., 2018). Thus, a policymaker might be interested in the optimal response to such shocks, or how optimal policy more generally depends on risk and the distribution of shocks (Evans et al., 2016).

Finally, in some cases a first-order approximation might simply be inaccurate, and a higher-order approximation is preferred. While numerical methods for handling such problems are already well established (as noted earlier), it can be helpful to gain further insight by analyzing the polynomial representation that is equivalent to the higher-order approximation. This includes reformulating the objective function to identify variables that are present in the objective (i.e. the objectives of policy), under what conditions those variables have non-zero targets, to identify the shocks that present trade-offs for policy, and how higher-order terms matter too. We now analyze such considerations using a prototypical example of optimal monetary policy in the New Keynesian (NK) model.

III. An NK model with Costly Wage and Price Adjustment

We study a standard NK model with costly wage and price adjustment similar to Kim and Ruge-Murcia (2009). The policymaker (think of a central bank) implements monetary policy optimally by choosing a sequence of nominal interest rates to maximize household

welfare under commitment. A brief description of the economy and optimal monetary policy follows.

We assume a continuum of identical households and firms. Each household, indexed by $h \in [0, 1]$, chooses their optimal consumption bundle of differentiated goods (C_{it}^h for $i \in [0, 1]$), labor supply (H_t^h), and a state-contingent portfolio of one-period securities ($\mathcal{Q}_{t+1}^h$) to maximize lifetime utility $E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} (U(C_t^h; Q_t) - V(H_t^h)) \right]$, subject to the sequence of flow budget constraints that hold for $t \geq t_0$:

$$\int_0^1 P_{it} C_{it}^h di + E_t \left[\mathcal{M}_{t,t+1} \mathcal{Q}_{t+1}^h \right] \leq \mathcal{Q}_t^h + W_t^h H_t^h (1 - \tau^w) (1 - \Phi_t^h) - T_t + \Xi_t$$

where felicity from consumption, $U(C_t^h; Q_t) := \frac{Q_t C_t^{h^{1-\gamma}}}{1-\gamma}$, is obtained from the consumption bundle ($C_t^h := \left(\int_0^1 (C_{it}^h)^{\frac{1}{\varepsilon}} di \right)^\varepsilon$), $\gamma > 0$ is the coefficient of relative consumption risk aversion (hereafter risk aversion) and $\varepsilon > 1$ determines the elasticity of substitution between goods given by $\varepsilon/(\varepsilon - 1)$, and P_{it} denotes the price of consumption good type i . Q_t is a consumption demand (hereafter demand) shock and $V(H_t^h) := \frac{H_t^{h^{1+\eta}}}{1+\eta}$ the disutility from supplying labor with its curvature governed by $\eta \geq 0$. Ξ_t denotes any rents accrued from shares held in monopolistically competitive firms, T_t is the nominal value of lump-sum taxes paid and τ^w is a proportional tax (or subsidy) on wage income (net of adjustment costs) that can be set to eliminate the steady state distortion associated with monopolistic labor supply (discussed further below). We assume perfect consumption insurance across households with a unique stochastic discount factor ($\mathcal{M}_{t,t+1}$) between periods t and $t + 1$. The policy instrument for the central bank is the risk-free nominal interest rate, i_t^b , that can be used to price the state-contingent portfolio.

Households supply labor monopolistically and choose their optimal wage, W_t^h , subject to the demand for their labor, $H_t^h = \left(\frac{W_t^h}{W_t} \right)^{-\frac{\theta_t}{\theta_t-1}} H_t$, where θ_t is a time-varying ‘wage-markup’ shock.²⁰ Following Kim and Ruge-Murcia (2009, 2011), we allow for asymmetry in wage adjustment costs, $W_t^h \Phi_t^h$, and assume the adjustment cost function $\Phi_t^h = \frac{\phi}{\psi^2} \left(\exp \left(-\psi \left(\frac{W_t^h}{W_{t-1}^h} - 1 \right) \right) + \psi \left(\frac{W_t^h}{W_{t-1}^h} - 1 \right) - 1 \right)$ is of the Linex class where $\phi \geq 0$ governs the magnitude of wage adjustment costs, and ψ the degree of asymmetry in them.²¹ This asymmetry will prove to be quantitatively important, and allows for the idea that it is more costly to lower nominal wages than to increase them ($\psi > 0$).²² In an equilibrium

²⁰Like Erceg et. al (2000) we assume a representative ‘employment agency’ that combines imperfectly substitutable labor using a Dixit-Stiglitz aggregator and supplies labor hours in the proportions firms would choose.

²¹Note as $\psi \rightarrow 0$ the cost function approaches the symmetric (quadratic) adjustment cost case, while as ψ increases it becomes more costly to reduce nominal wage inflation than to increase it.

²²Downward nominal wage rigidity is commonly used across a range of macro models (see for example Eggertsson,

where all households are identical, household optimization implies:

$$(8) \quad \lambda_t = \beta E_t \left[\lambda_{t+1} \left(1 + i_t^b \right) \right]$$

$$(9) \quad \frac{1 - \Phi_t}{\theta_t - 1} + \frac{\partial \Phi_t}{\partial \Pi_t^w} \Pi_t^w = \frac{V_H(H_t)}{\lambda_t (1 - \tau^w)} \frac{\theta_t}{\theta_t - 1} (W_t)^{-1} + E_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{H_{t+1}}{H_t} \frac{\partial \Phi_{t+1}}{\partial \Pi_{t+1}^w} (\Pi_{t+1}^w)^2 \right]$$

where $\lambda_t := \frac{U_c(C_t; Q_t)}{P_t}$ is the marginal utility of consumption deflated by the price level, $W_t := \left(\int_0^1 W_t^{h^{1/(1-\theta_t)}} \right)^{1-\theta_t}$ is the aggregate wage index, $\Pi_t^w := W_t/W_{t-1}$ is wage inflation and wage adjustment costs are the same across all households $\Phi_t^h = \Phi_t \forall h \in [0, 1]$.

Firms, indexed by $i \in [0, 1]$, produce a differentiated good (Y_{it}) using a linear production technology, $Y_{it} = A_t H_{it}$, where H_{it} is the quantity of labor employed by firm i and A_t is a common productivity shock. Minimizing costs, which in real marginal terms are $\frac{S_{it}}{P_t} := \frac{W_{it}}{P_t} \frac{1}{A_t}$, firms compete monopolistically and set their optimal price to maximize profit $E_{t_0} \left[\sum_{t \geq t_0} \mathcal{M}_{t_0,t} (P_{it} - (1 - \varsigma) S_{it} \Upsilon_t - \Phi_{it}^\pi P_{it}) Y_{it} \right]$ subject to the demand for their goods, $Y_{it} = \left(\frac{P_{it}}{P_t} \right)^{-\frac{\varepsilon}{\varepsilon-1}} Y_t$, and quadratic (Rotemberg) price adjustment costs $\Phi_{it}^\pi := \frac{\xi}{2} \left(\frac{P_{it}}{P_{it-1}} - 1 \right)^2$ where $\xi \geq 0$. We could also allow for asymmetry in price adjustment costs, but as there is less evidence to support this assumption we abstract from it here.²³ Υ_t is an exogenous cost-push shock and ς a production subsidy that eliminates the steady state distortion associated with monopolistic competition among firms. Optimal pricing in a symmetric equilibrium where all firms are identical implies:

$$(10) \quad \frac{1 - \Phi_t^\pi}{\varepsilon - 1} + \Pi_t \frac{\partial \Phi_t^\pi}{\partial \Pi_t} = \frac{\varepsilon (1 - \varsigma)}{\varepsilon - 1} \left(\frac{S_t}{P_t} \Upsilon_t \right) + \beta E_t \left[\frac{\lambda_{t+1} Y_{t+1}}{\lambda_t Y_t} \frac{\partial \Phi_{t+1}^\pi}{\partial \Pi_{t+1}} \Pi_{t+1}^2 \right]$$

where $\Pi_t := P_t/P_{t-1}$ is price inflation and $\Phi_{it}^\pi = \Phi_t^\pi \forall i \in [0, 1]$. All markets, goods, labor, and state-contingent financial claims clear and household transversality conditions are satisfied. The aggregate production function and resource constraint are given by:

$$(11) \quad Y_t = A_t H_t$$

$$(12) \quad Y_t = C_t + \Phi_t^\pi Y_t + \frac{W_t}{P_t} \Phi_t H_t$$

The optimal nonlinear Ramsey monetary policy (without approximation) in a symmetric equilibrium is given by sequences $\left\{ C_t^h = C_t, H_t^h = H_t, Y_t, \Pi_t, \Pi_t^w, \frac{W_t}{P_t}, i_t^b \right\}_{t \geq t_0}$ that maximize lifetime household utility (welfare) subject to the decentralized equilibrium conditions (8) to (12), and where nominal wage inflation is given by $\Pi_t^w = \frac{W_t/P_t}{W_{t-1}/P_{t-1}} \Pi_t$. We now turn to characterizing the LQ ($n = 1$) and QC ($n = 2$) approximations of this policy.

Mehrotra and Robbins (2019) and Schmitt-Grohé and Uribe (2017)).

²³See Kim and Ruge-Murcia (2009).

IV. LQ and QC Approximations of Optimal Monetary Policy

We focus on the equilibrium where the steady state wage income tax and production subsidy are chosen to eliminate the inefficiently low level of output associated with monopolistic competition in labor supply and production. Thus, the central bank has no incentive to use monetary policy to offset the steady state effects of these distortions. An approach commonly applied with LQ,²⁴ it facilitates the most transparent comparison of the LQ and QC approximations. However, nothing in our earlier theoretical contribution requires these assumptions. Indeed, our earlier contribution shows how to take a higher-order approximation of optimal policy when the steady state is inefficient.

We first consider a version of the model where consumption risk aversion is set to 1 (log utility from consumption, denoted LU), only price adjustment is costly, and only productivity and cost-push shocks perturb the economy. This takes us to a model that has the same implications for optimal monetary policy design as the canonical NK model (Nisticó, 2007; Clarida, Gali and Gertler, 1999).

Even in this very simple model, there are clear differences between the two approximations as Proposition 2 shows. Reporting the LQ ($n = 1$) and QC ($n = 2$) approximations of optimal policy, only the non-highlighted terms (in black) are present in the LQ approximation; with QC, both non-highlighted and highlighted terms (in blue) are present.²⁵

PROPOSITION 2: *The QC approximation of optimal policy around an efficient steady state assuming log utility ($\gamma \rightarrow 1$), no wage adjustment costs ($\phi = 0$) and only productivity and cost-push shocks are present ($\text{var}(a_t) > 0$, $\text{var}(\mu_t) > 0$) is:*

$$(13) \quad \mathcal{L}_{t_0}^{QC-LU} = \min_{\{\hat{i}_t^b, x_t, \pi_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(\xi \frac{\pi_t^2}{2} + (1+\eta) \frac{x_t^2}{2} + \xi \frac{\pi_t^3}{2} + (1+\eta)^2 \frac{x_t^3}{3!} \right) \right]$$

subject to the 2nd-order expansion of the constraint set:

$$(14) \quad \pi_t + \frac{3}{2} \pi_t^2 = \kappa s_t^g + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right]$$

$$(15) \quad E_t [r_{t+1}^g] = E_t \left[\Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g - \Delta x_{t+1})^2 - \frac{\xi}{2} \Delta \pi_{t+1}^2 \right]$$

where the real marginal cost and interest rate gaps are given by:

$$s_t^g := (1+\eta) x_t + \mu_t + \frac{1}{2} ((1+\eta) x_t + \mu_t)^2 - \frac{\xi}{2} \pi_t^2, \quad r_{t+1}^g := \hat{i}_t^b - r_{t+1}^{bn} - \pi_{t+1}$$

²⁴ Assuming an efficient steady state is common with LQ (see for example, Garga and Singh (2021) and Ravenna and Walsh (2011)).

²⁵ Note we now drop approximation remainders for brevity.

and the flexible-price equilibrium natural interest rate and output by $r_{t+1}^{bn} := \Delta y_{t+1}^n$ and $y_t^n := a_t$ and $\kappa := \xi^{-1}(\varepsilon - 1)^{-1}$ and where the variances of demand and wage mark-up shocks are zero ($\text{var}(q_t) = \text{var}(\hat{\theta}_t) = 0$). While the QC approximation includes all terms, the LQ approximation includes only quadratic terms in the objective and linear terms in the constraints.

Proposition 2 is derived using variables that are measured in terms of deviations from their flexible-price equilibrium (natural) values (e.g. the output gap is $x_t := y_t - y_t^n$ where y_t^n is natural output), and lower case is used to denote each variables' log deviation from steady state (e.g. $\pi_t := \ln \Pi_t / \bar{\Pi}$).²⁶ The constraints are given by the 2nd-order expansions of the New Keynesian Phillips Curve (NKPC, 14) and New Keynesian IS Curve (NKIS, 15), where r_t^{bn} is the natural interest rate and $E_t[r_{t+1}^g]$ the expected real interest rate gap.²⁷

While the definitions of the flexible-price equilibrium in the two approximating problems are identical, there are important differences in the loss functions and the constraints that affect optimal monetary policy design. This follows from the additional 3rd-order terms in the loss under the QC approximation ($\xi \pi_t^3 / 2 + (1 + \eta)^2 x_t^3 / 3!$) that result in asymmetry.²⁸ These terms imply the central bank is *more averse* to positive deviations in inflation and the output gap than negative deviations under QC.²⁹ With log utility, all second- and higher-order terms in the log deviations in consumption from steady state drop out of any higher-order approximation and so there are no higher-order effects through which consumption volatility (or skewness) matters.³⁰

This result does not hold more generally though. With consumption risk aversion different from one, whether the central bank prefers expansions or contractions will in general depend on the degree of consumption risk aversion, the degree of curvature in the disutility of labor supply and any asymmetries induced by the constraint set, as we discuss below.

The next corollary shows that the Divine Coincidence does not hold for productivity shocks when cost-push shocks have a positive variance under QC. Under LQ, the Divine coincidence always holds for productivity shocks regardless of whether the variance of cost-push shocks is positive or zero.

²⁶Where the lower case is already used for a variable label, we append a hat (e.g. $\hat{i}_t^b := \ln(1 + i_t^b) - \ln(1 + \bar{i})$).

²⁷That is, the expected deviation between the real rate of interest chosen by the central bank and the natural interest rate.

²⁸Note such asymmetries are not confined to this example. Using the properties of even and odd functions, one can show these results apply more generally when $n > 1$.

²⁹A similar result is also found in Benigno and Rossi (2021), who study a second-order approximation with log utility from consumption and Calvo pricing.

³⁰To see this formally, just evaluate a third-order expansion of $\log(C_t)$ and note the terms c_t^2 and c_t^3 drop out of the approximation.

COROLLARY 1: *When cost-push shocks can perturb the economy in the future (i.e. $\text{var}(\mu_t) > 0$), the policy $\{x_t = 0, \pi_t = 0, i_t^b = E_t[r_{t+1}^{bn}]\}_{t \geq t_0}$ is optimal under LQ in response to any realized productivity shock a_{t_0} . However, this is not an optimal policy under QC.*

Corollary 1's results for LQ should not be surprising. Benigno and Woodford (2004) have previously showed that under a first-order approximation with an efficient steady state, the optimal policy response to a productivity shock entails full price stability and a stabilized output gap at zero in all periods (the so-called Divine Coincidence). Indeed, this result can also be proven directly from the nonlinear optimality conditions of the policymaker's problem.³¹ What is less well known, with a higher-order approximation, is that the potential for cost-push shocks to arrive in any future period, $t > t_0$, is sufficient to break the optimality of full price stability in response to a productivity shock realized in period t_0 , even *before* any cost-push shock is realized.³² That is, the potential for cost-push shocks to perturb the economy in the future, presents a trade-off for monetary policy today.

This mechanism is only present in the QC approximation, and is not present under LQ. It arises from the differences in the NKPC under the two approximations. Solving the NKPC (14) forwards in each case:

$$\begin{aligned} \text{NKPC}^{LQ-LU}: \pi_t &= \kappa E_t \left[\sum_{\tau \geq 0} \beta^\tau s_{t+\tau}^{g,LQ} \right], & s_{t+\tau}^{g,LQ} &:= (1 + \eta) x_{t+\tau} + \mu_{t+\tau} \\ \text{NKPC}^{QC-LU}: \pi_t + \frac{3}{2} \pi_t^2 &= \kappa E_t \left[\sum_{\tau \geq 0} \beta^\tau \left(s_{t+\tau}^{g,QC} \right) \right] \\ s_{t+\tau}^{g,QC} &:= (1 + \eta) x_{t+\tau} + \mu_{t+\tau} + \underbrace{\frac{1}{2} ((1 + \eta) x_{t+\tau} + \mu_{t+\tau})^2 - \frac{\xi}{2} \Delta \pi_{t+\tau}^2}_{\text{Future cost uncertainty}} \end{aligned}$$

Here, we see that the only variance that presents a trade-off for policy is that for cost-push shocks, which implies $E_t[\mu_{t+\tau}]^2 > 0$ for any $\tau > 0$. This variance generates future cost uncertainty and shifts the intercept in the NKPC under QC, creating a trade-off between stabilizing inflation and the output gap for productivity shocks, as firms choose higher prices anticipating the potential for changes in their costs in the future. This breaks the Divine Coincidence as the central bank must choose between full price stability (and thus a non-zero output gap) or closing the output gap (resulting in non-zero inflation). With certainty equivalence in the LQ approximation, this mechanism is not present and the

³¹We are grateful to a referee for drawing our attention to this. Similar results are found in Khan, King and Wolman (2003).

³²Other mechanisms that can break the optimality of price stability at the first-order include an inefficient steady state due to government purchases and inefficient taxation (Benigno and Woodford, 2005) or due to household heterogeneity and a liquidity-insurance motive (Bilbiie and Ragot, 2021).

Divine Coincidence always holds for productivity shocks.

Proposition 2 and Corollary 1 are derived assuming taxes and subsidies for labor income and goods imply an efficient steady state. If this is not the case, and the steady state is inefficient, then there are additional linear terms in the QC and LQ approximations of household welfare. From Proposition 1, these terms must then be substituted out using a 3rd-order (2nd-order) approximation of the constraints under QC (LQ). For brevity, we do not explore that extension here, but examples of this approach applied with LQ include Benigno and Woodford (2005) and Bilbiie and Ragot (2021).

A. LQ and QC Approximations in the General NK Economy

Having established simple results for the canonical representation of the NK model, we now additionally allow for a richer set of shocks (wage markup ($\text{var}(\hat{\theta}_t) > 0$) and demand ($\text{var}(q_t) > 0$)), for more (or less) risk averse households with CRRA utility ($\gamma \neq 1$), and explore the role of asymmetry in the constraints through costly nominal wage adjustment ($\phi > 0, \psi \neq 0$). To highlight the differences between the QC and LQ approximations, again non-highlighted terms are present in both approximations while those in blue are also present in the QC approximation only.

PROPOSITION 3: *The QC approximation of the optimal Ramsey problem around the efficient steady state is:*

$$\min_{\{\hat{w}_t^b, x_t, \pi_t, \pi_t^w, w_t^g\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} (L_t^{QC}) \right]$$

$$\begin{aligned} L_t^{QC} := & (\gamma + \eta) \left(1 + \omega_t^{QC} \right) \frac{x_t^2}{2} + \xi \left(1 + \omega_t^{QC} \right) \frac{\pi_t^2}{2} - \left((\gamma - 1)^2 - (1 + \eta)^2 \right) \frac{1}{3!} x_t^3 \\ & - \xi (\gamma - 1) \frac{\pi_t^2 x_t}{2} + \xi \frac{\pi_t^3}{2} + \frac{1}{2} \phi \left(\left(1 + \omega_t^{QC} \right) \pi_t^{w^2} + \left(1 - \frac{1}{3} \psi \right) \pi_t^{w^3} + \pi_t^{w^2} w_t^g - (\gamma - 1) \pi_t^{w^2} x_t \right) \end{aligned}$$

with state-contingent weight $\omega_t^{QC} := \frac{1+\eta}{\gamma+\eta} (q_t - (\gamma - 1) a_t)$, and where the minimization is subject to the 2nd-order expansion of the NKWPC:

$$\pi_t^w + (3 - \psi) \frac{\pi_t^{w^2}}{2} = \frac{1}{\phi(\theta - 1)} (J_t) + \beta E_t \left[\pi_{t+1}^w + (3 - \psi) \frac{\pi_{t+1}^{w^2}}{2} \right] + \beta E_t [\Xi_{t+1}^w \pi_{t+1}^w]$$

the marginal disutility of supplying labor:

$$\begin{aligned} J_t := & (\gamma + \eta) x_t - w_t^g + \frac{1}{2} ((\gamma + \eta) x_t - w_t^g)^2 + \left(\frac{1}{\theta - 1} \right) \hat{\theta}_t (w_t^g - (\gamma + \eta) x_t) \\ & - \gamma \xi \frac{\pi_t^2}{2} - (\gamma - 1) \phi \frac{\pi_t^{w^2}}{2} + \hat{\theta}_t - \frac{1 + \theta}{\theta - 1} \frac{\hat{\theta}_t^2}{2} \end{aligned}$$

the 2nd-order expansions of the NKPC and NKIS curves:

$$\begin{aligned}\pi_t + \frac{3}{2}\pi_t^2 &= \kappa s_t^g + \beta E_t \left[\pi_{t+1} + \frac{3}{2}\pi_{t+1}^2 \right] + \beta E_t \left[\Xi_{t+1}^\pi \pi_{t+1} \right] \\ E_t \left[r_{t+1}^g \right] &= E_t \left[\gamma \Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g - \gamma \Delta x_{t+1})^2 - \gamma \frac{\xi}{2} \Delta \pi_{t+1}^2 - \gamma \frac{\phi}{2} \Delta \pi_{t+1}^w \right]\end{aligned}$$

with real marginal cost, interest-rate, wage and output gaps and wage inflation given by:

$$\begin{aligned}s_t^g &:= w_t^g + \mu_t + \frac{1}{2} (w_t^g + \mu_t)^2 - \xi \frac{\pi_t^2}{2}, & r_{t+1}^g &:= \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn} \\ w_t^g &:= w_t - w_t^n, & x_t &:= y_t - y_t^n, & \pi_t^w &:= \Delta w_t + \pi_t\end{aligned}$$

the flexible price and wage equilibrium is:

$$y_t^n := \frac{q_t}{\gamma + \eta} + \frac{1 + \eta}{\gamma + \eta} a_t, \quad \hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}, \quad w_t^n := a_t$$

and given covariance terms:

$$\begin{aligned}\Xi_{t+1}^w &:= \Delta y_{t+1}^n - \Delta w_{t+1}^n - (\gamma - 1) \Delta x_{t+1} - \left(\hat{r}_{t+1}^{bn} + \pi_{t+1} - \pi_{t+1}^w \right) \\ \Xi_{t+1}^\pi &:= \Delta y_{t+1}^n - \hat{r}_{t+1}^{bn} - (\gamma - 1) \Delta x_{t+1}\end{aligned}$$

While the QC approximation includes all terms, the LQ approximation includes only quadratic terms in the objective and linear terms in the constraints.

Allowing for CRRA utility ($\gamma \neq 1$) changes the curvature in both the loss function and the constraints. In the loss function it provides additional asymmetry through the 3rd-order terms $(-\gamma - 1)^2 \frac{1}{3!} x_t^3 - \xi (\gamma - 1) \frac{\pi_t^2 x_t}{2} - \phi (\gamma - 1) \frac{\pi_t^w x_t}{2})$, which imply the central bank becomes *more* averse to negative than positive deviations in the output gap as γ increases. In the constraints, increasing γ amplifies the role of price- and wage-inflation risk premia in the transmission of monetary policy through the NKIS curve 2nd-order terms $(-\gamma \frac{\xi}{2} \Delta \pi_{t+1}^2 - \gamma \frac{\phi}{2} \Delta \pi_{t+1}^w)$. These imply a higher γ increases the price of a risk free government bond (spanned by the portfolio of one-period securities), reducing the real interest rate gap.

Perhaps more surprisingly, the QC approximation implies a representation of the loss function where the shocks perturb the weights on the 2nd-order terms in inflation and the output gap ($\omega_t^{QC} = \frac{1+\eta}{\gamma+\eta} (q_t - (\gamma - 1) a_t)$), which are increasing (decreasing) in demand (productivity) shocks. As we show below, it implies that a central bank's preferences vary over the business cycle as it optimally trade-offs skewness (3rd-order variation) against volatility (2nd-order variation). This mechanism is, of course, absent under LQ.

Finally, costly wage adjustment implies a second Phillips curve in wage inflation (the NKWPC), linking current wage inflation to the current disutility of supplying labor and expected future wage inflation. It also implies that wage inflation enters the loss function as a

target variable in its own right through the terms $(\frac{1}{2}\phi \left((1 + \omega_t^{QC}) \pi_t^w + (1 - \frac{1}{3}\psi) \pi_t^w + \pi_t^w w_t^g \right))$. Thus, when downward asymmetry (ψ) is large, the costs of reducing nominal wage inflation are higher than the costs of increasing it, as captured by the 3rd-order term $(1 - \frac{1}{3}\psi) \pi_t^w$.

V. Quantitative Analysis

How do these considerations affect optimal monetary policy design quantitatively? To parameterize the approximations and compare them, we use quarterly US data from 1964:II to 2006:II and estimate the more general (2nd-order) model via Simulated Method of Moments as per Kim and Ruge-Murcia (2009).³³ When estimating the model, we do not impose that the Federal Reserve always acted optimally. Instead, its empirical policy function is summarized by a simple Taylor rule of the class:

$$\widehat{i}_t^b = \rho_i^b \widehat{i}_{t-1}^b + \left(1 - \rho_i^b\right) (\rho_\pi \pi_t + \rho_w \pi_t^w + \rho_x x_t + \rho_r \Delta y_t)$$

This class of interest-rate rule is appealing because it responds to the target variables in the welfare-theoretic loss derived earlier, and provides a good summary of policy empirically.³⁴ Table 1 reports the estimates, which are similar to those reported in previous literature including Kim and Ruge-Murcia (2009). We now use these estimates to quantify the differences in optimal policy between the LQ and QC approximations.³⁵

TABLE 1—SMM PARAMETER ESTIMATES

Parameter	Description	Parameter	Description
γ	2.52 Risk aversion	ρ_r	0.69 Output growth weight
η	1.19 Labor disutility	ρ_i^b	0.23 Interest rate smoothing
ξ	37.8 Price curvature	ρ_a	0.92 Productivity persistence
ϕ	280.4 Wage curvature	ρ_z	0.99 Demand persistence
ψ	3844.4 Wage asymmetry	ρ_θ	0.49 Wage-markup persistence
ε	1.1 Goods substitutability	ρ_μ	0.00 Cost-push persistence
θ	1.4 Labor substitutability	σ_a	0.0132 Stdev. productivity
β	0.997 Discount factor	σ_d	0.0106 Stdev. demand
ρ_π	1.67 Price inflation weight	σ_θ	0.0298 Stdev. wage-markup
ρ_w	0.00 Wage inflation weight	σ_μ	0.0008 Stdev. cost-push
ρ_x	1.40 Output gap weight		

Note: The parameters determining the markup over goods (ε), and labor types (θ), and the discount factor (β) are calibrated as per Kim and Ruge-Murcia (2009). All other parameters are estimated using Simulation Method of Moments (see online Appendix D).

³³Different to Kim and Ruge-Murcia, our model allows for cost-push and wage-markup shocks. This permits a greater role for supply-side shocks in generating business cycle fluctuations and is crucial in generating trade-offs for monetary policy. These shocks are also common in larger estimated DSGE models such as Smets and Wouters (2007).

³⁴This class of rules has been studied by Coibion and Gorodnichenko (2012), for example.

³⁵The exception is the Taylor rule parameters, which are only used to compute welfare under different policy regimes, for historical policy counterfactuals and as a benchmark for comparison when assessing accuracy.

Figure 1 displays the combinations of price inflation and the output gap (left-hand panel) and wage inflation and the output gap (right-hand panel) that the central bank is indifferent to. As one traces over the indifference curves under LQ (the solid black line), the central bank's optimal trade-off is symmetric. With QC (the dotted blue line), there is no such symmetry. The regions where the indifference curves are flatter are those where the central bank places *greater marginal value* on stabilizing the output gap relative to price (or wage) inflation. Conversely, regions where the indifference curves are steeper show where the central bank is less willing to trade-off price (or wage) inflation volatility for output gap volatility. The asymmetry is most pronounced in the indifference curves for wage inflation, reflecting the downward asymmetry estimated in wage adjustment costs.

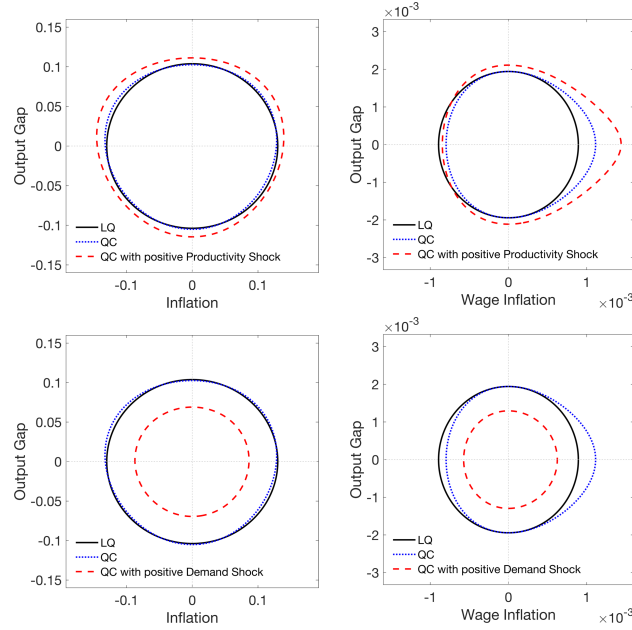


FIGURE 1. LQ AND QC INDIFFERENCE CURVES

Notes: These curves show combinations of price and wage inflation and the output gap where the central bank is indifferent under LQ and QC for the same fixed welfare level. Points closer to the origin denote higher welfare.

Figure 1 also shows how the indifference curves shift in response to one standard deviation changes in productivity and demand (the dashed red lines). While these shifts do not happen with LQ, under QC a positive productivity shock reduces the weight on the 2nd-order terms in inflation and the output gap, and the indifference curves expands outwards from the origin. As the shock dissipates, they contract to their original position. Demand shocks lead to similar shifts, but in the opposite direction.

One might conjecture these results require asymmetry in wage adjustment costs. Figure 2 shows that this is not so. It displays regions of the parameter space where the central bank prefers both positive deviations in inflation and the output gap relative to negative devia-

tions, depending on household risk aversion (γ), the disutility of labor (η) and symmetric price adjustment costs (ξ) and assuming costless nominal wage adjustment ($\phi = 0$).

The left-hand panel shows increasing household risk aversion changes the combinations of price-adjustment costs and labor disutility where the central bank still prefers positive deviations in inflation and the output gap than negative (the line denoting symmetry shifts down and to the right). As risk aversion increases, low price adjustment costs and a high disutility of labor are required for the central bank to prefer negative deviations in inflation and the output gap. The right-hand panel shows the effect of changing the curvature in the disutility of labor supply. Increasing the latter implies high risk aversion and price-adjustment costs are required for positive deviations to be preferred (the line of symmetry shifts up and to the left).

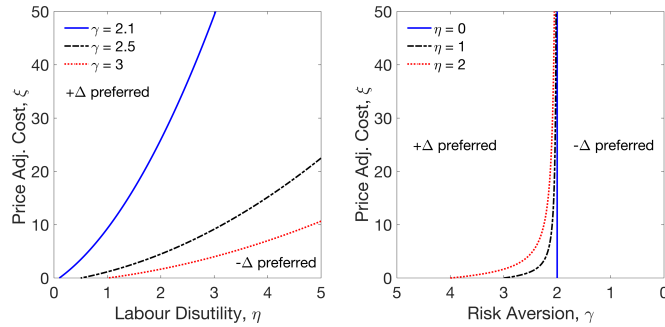


FIGURE 2. ASYMMETRY IN PREFERENCES UNDER QC

Notes: “+ Δ ” denotes where the central bank prefers positive deviations in inflation and output gaps relative to equally-sized negative ones with $\phi = 0$. “- Δ ” denotes regions where negative deviations are preferred. The black, red and blue lines denote the lines of symmetry (equal weighting) at the parameter values in the top left of each subfigure.

A. Optimal Policy

We next report how optimal policy changes under the two approximations in response to different shocks. Here, we follow Andreasen, Fernández-Villaverde and Rubio-Ramírez (2018) and solve the FOC of the QC model numerically using the pruned state space. Figures 3 and 4 show impulse response functions (IRFs) to 1 and 2 standard deviation shocks to productivity and wage-markups. With LQ there is no account for the asymmetry in wage adjustment costs and so larger falls in the nominal wage occur in response to negative productivity or wage-markup shocks. Both call for an optimal fall in the nominal wage though for different reasons – with negative productivity because the marginal product of labor has fallen, with a lower mark-up because households demand less compensation for the same supply of labour. With QC, declines in the nominal wage are perceived as more costly (relative to LQ) and the central bank chooses a policy that avoids substantial wage deflation in both cases. Indeed, when the shocks are large, these effects can even

flip the sign of the optimal output gap, wage inflation or interest rate responses across the two approximations. These differences are not only confined to negative shocks, significant differences are also observed for positive shocks too.

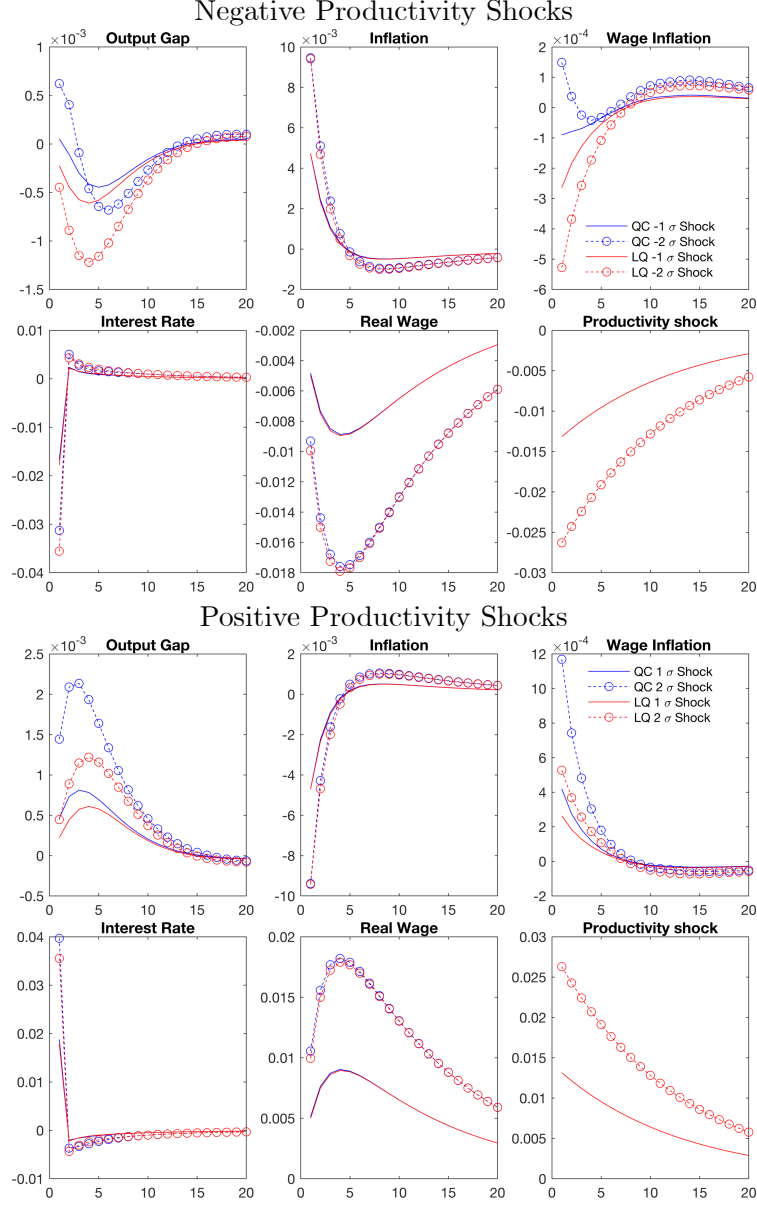


FIGURE 3. IRFs FROM THE LQ AND QC SOLUTIONS TO A NEGATIVE AND POSITIVE PRODUCTIVITY SHOCK

Notes: 1σ and 2σ denote 1 and 2 standard deviation shocks.

Figures D1–D2 in the online Appendix report the IRFs for demand and cost-push shocks. For them, the quantitative differences in the optimal responses across the two approximations are smaller with low volatility and persistence estimated for cost-push shocks.

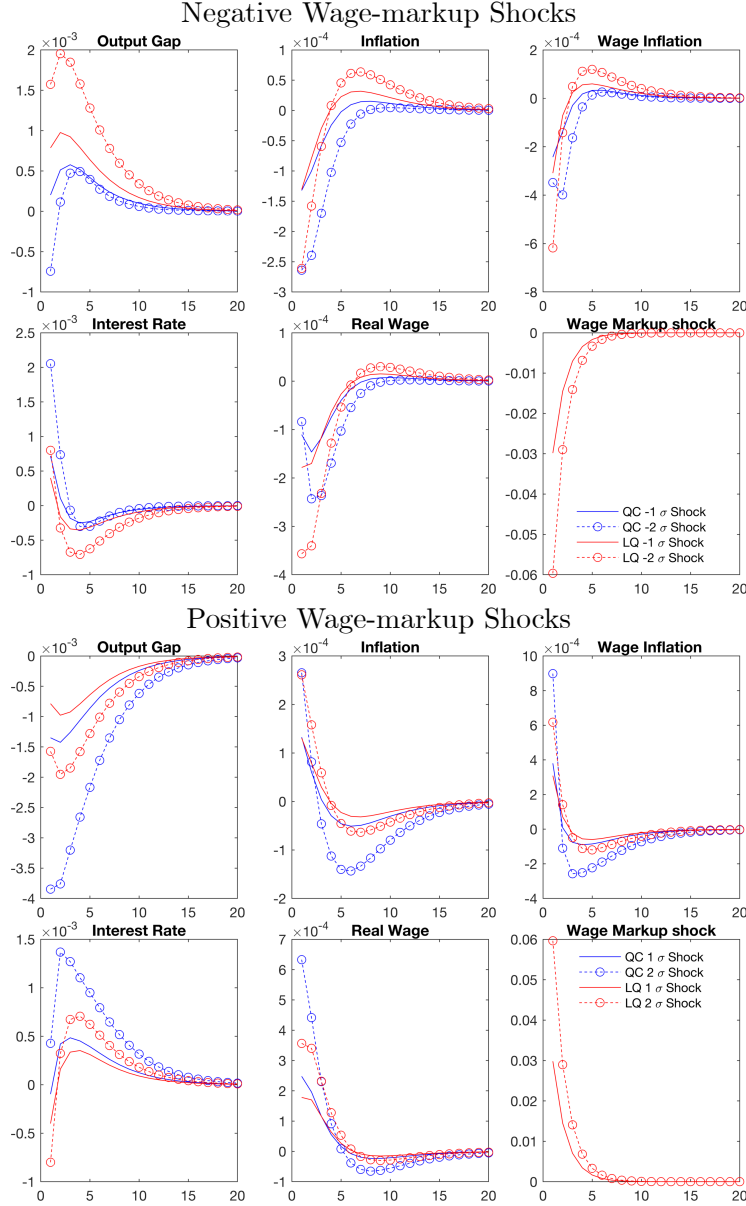


FIGURE 4. IRFs FROM THE LQ AND QC SOLUTIONS TO A NEGATIVE AND POSITIVE WAGE-MARKUP SHOCKS

Notes: 1σ and 2σ denote 1 and 2 standard deviation shocks.

Figure 5 shows the simulated marginal distributions of wage- and price-inflation, the output gap and the nominal interest rate under the two approximations. While the distributions are similar for price inflation (albeit with a small increase in the mean under QC), there are notable differences for the other variables. The distributions for wage inflation and the output gap are positively skewed with less mass at negative values under QC, as the central bank optimally avoids wage deflations and output gap contractions. Reflecting the second-order risk premia in the transmission of policy, the distribution for the nominal interest rate also exhibits positive skewness. With LQ, all distributions are symmetric.

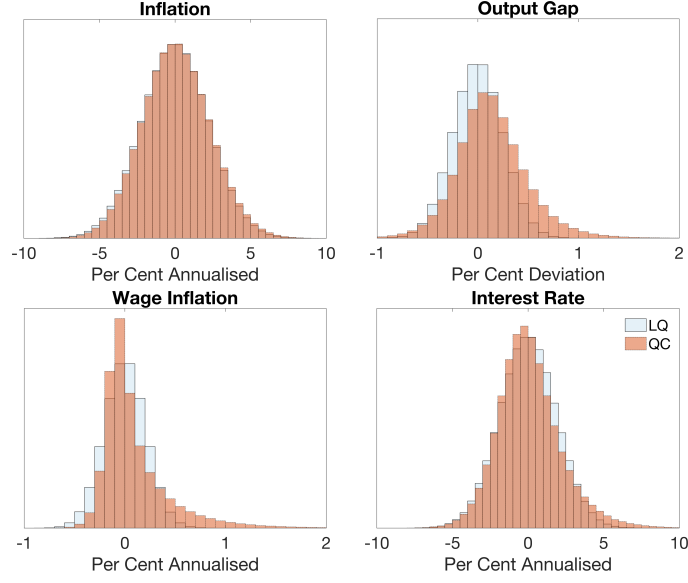


FIGURE 5. SIMULATED MARGINAL HISTOGRAMS

Figures 6 and 7 compare the simulated joint distributions. With LQ, we again observe symmetry with positive comovement in nominal interest rates and price inflation, and between the output gap and wage inflation. In contrast, with QC the joint distributions are curved. There is negative truncation of the joint distribution of wage inflation and the output gap, and a nonlinear relationship between price inflation and the nominal interest rate, reflecting the state-contingent trade-off between stabilizing price- and wage-inflation under QC.

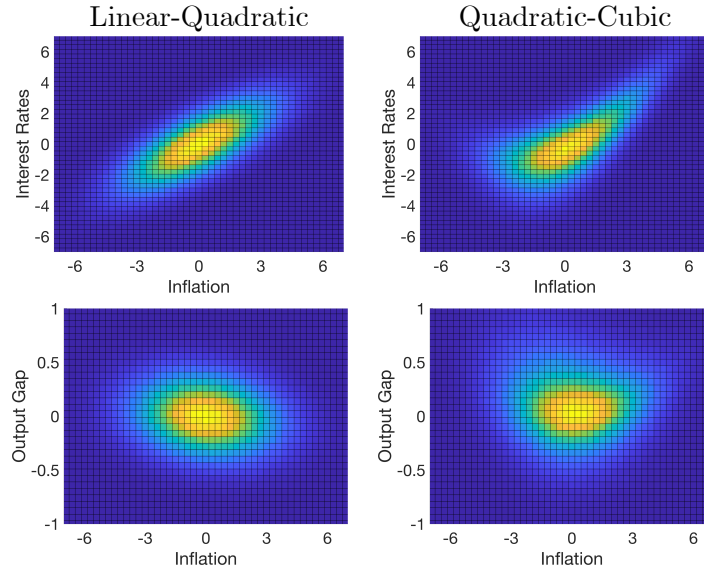


FIGURE 6. SIMULATED JOINT HISTOGRAMS

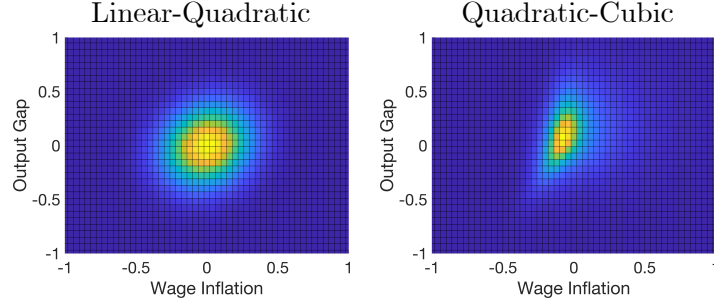


FIGURE 7. SIMULATED JOINT HISTOGRAMS

B. Welfare and Taylor Rules

We next discuss the welfare implications of using a Taylor rule as opposed to optimal policy. We compute the change in the unconditional mean of welfare under the LQ and QC loss functions and where the Taylor rule coefficients are those estimated in Table 1. Optimal policy is either optimal LQ or QC policy. The welfare change in each case is calculated using the mean of the ergodic distribution for welfare, measured relative to the deterministic steady state and reported in annualized consumption equivalent values.

Table 2 shows the aggregate results for all shocks and for each shock individually.³⁶ With all shocks, the welfare losses of using the estimated Taylor rule are lower when using the LQ, rather than QC, welfare measure (rows 1 and 4). Also, mean welfare is higher under optimal QC policy than under optimal LQ policy (rows 2 and 5). Together, they imply the welfare losses from using a Taylor rule are sizable under QC (row 6), but small with LQ (row 3). Relative to the steady state, households are willing to give up 0.05% (0.32%) of annual consumption when comparing a Taylor rule to optimal LQ (QC) policy.

TABLE 2—A TAYLOR RULE, OPTIMAL POLICY AND WELFARE

LQ welfare metric	All shocks	Demand	Productivity	Cost-push	Wage Markup
Taylor rule	-0.180	-0.004	-0.168	0.000	-0.009
Optimal LQ policy	-0.128	0.000	-0.123	0.000	-0.005
Implied welfare loss	0.052	0.004	0.045	0.000	0.005
QC welfare metric					
Taylor rule	-0.436	-0.002	-0.107	0.000	-0.112
Optimal QC policy	-0.112	0.000	-0.114	0.000	-0.007
Implied welfare loss	0.324	0.002	-0.006	0.000	0.105

Notes: Taylor rule reports the change in welfare (relative to the deterministic steady state) when policy follows the empirical Taylor rule as per the estimates in Table 1. Optimal LQ (QC) policy denotes optimal policy as computed under the Linear-Quadratic (Quadratic-Cubic) approximation. All numbers are annualized consumption equivalents.

³⁶We compute the steady-state (annualized) consumption subsidy that would equate mean welfare under each policy to welfare obtained in the deterministic steady state. For example, a welfare change of -1 reported in Table 2 would imply households are willing to forgo 1% of their annual consumption to avoid moving from the deterministic economy to the corresponding stochastic economy under a given policy regime. For columns 2–5, the estimated standard deviation for each shock is used as per Table 1, while all other shocks are set to zero. The small welfare effects of cost-push shocks reflects their small standard deviation.

C. Historical Counterfactuals

We now turn to historical counterfactuals for how the US economy might have evolved on the 1964:II to 2006:II sample, had the Federal Reserve implemented optimal QC policy, optimal LQ policy and when compared with the Taylor rule estimated for that period. Table 3 shows the mean deviation (relative to the steady state), standard deviation, 10th (P10) and 90th (P90) percentiles, and skewness of the output gap, price inflation, wage inflation and nominal interest rates under each policy counterfactual.

TABLE 3—HISTORICAL POLICY COUNTERFACTUALS: 1964:II TO 2006:II

Taylor rule	Mean Dev.	Std Dev.	P10	P90	Skew.
Output gap	0.09	0.25	-0.15	0.37	-0.64
Price Inflation	-0.66	2.26	1.42	6.71	0.48
Wage inflation	-0.80	1.21	2.70	5.77	1.28
Interest rate	-0.63	2.95	2.58	9.64	0.12
<u>Optimal LQ policy</u>					
Output gap	0.24	0.24	-0.17	0.47	-1.04
Price Inflation	0.12	1.84	2.50	6.84	0.40
Wage inflation	-0.02	0.18	4.29	4.62	2.05
Interest rate	-0.21	1.88	4.02	8.87	0.78
<u>Optimal QC policy</u>					
Output gap	0.33	0.18	0.14	0.57	0.62
Price Inflation	0.27	1.86	2.64	7.04	0.68
Wage inflation	0.13	0.27	4.44	4.85	2.74
Interest rate	-0.29	2.01	3.66	8.24	0.53

Notes: The counterfactuals are computed using a nonlinear filter on historical data and with all parameters set at their Table 1 values. Taylor rule denotes the empirical Taylor rule. Optimal LQ and QC policy counterfactuals are constructed using the filtered shocks and assuming the Federal Reserve had committed to optimal policy under each solution. Mean Dev. denotes the mean deviation from the deterministic steady state, Std Dev. is the standard deviation, P10 and P90 are the 10th and 90th percentiles of the historical counterfactual outcomes, and Skew. is skewness calculated using the standardized third moment. For further detail, see online Appendix D.

Both LQ and QC optimal policy result in higher means and lower standard deviations in the output gap, price inflation, wage inflation and the nominal interest rate when compared with the Taylor rule. However, while LQ focuses on stabilizing volatility only, the policy-maker under QC places additional weight on stabilizing contractions relative to expansions. The 10th and 90th percentiles, and skewness, for the output gap, price inflation and wage inflation are all higher under QC than LQ.

The potential for asymmetry to change the optimal response distributions of inflation and the output gap when moving to a higher-order approximation is also shown in Benigno and Rossi (2021). Studying a more recent sample for the US that includes the Great Recession (1995:I to 2019:III), they show a second-order approximation of optimal policy in a New Keynesian economy with Calvo pricing (and without nominal wage rigidity) results in a

lower standard deviation for the output gap relative to a first-order approximation, and a higher mean and skewness for price inflation. The mechanism driving asymmetry in their results is, however, quite different. With log utility from consumption implying the policymaker prefers contractions over expansions in their model, it is the dampening of the transmission of cost-push shocks to inflation with price dispersion under Calvo pricing, that results in an expansionary bias for optimal policy overall.

D. Accuracy

How accurate are the LQ and QC approximations? Table 4 presents the average Euler equation errors for each approximation under optimal policy, and when a Taylor rule is used. The errors are reported in a decimal log scale such that -3 indicates a \$1 in \$1,000 error when approximating the household consumption decision rule. The first column shows the average errors in the model with log utility from consumption, linear disutility of labor and where the only friction in the model is the quadratic price adjustment cost. We then introduce each source of additional curvature – beginning with non-unitary risk aversion (CRRA utility $\gamma > 1$), then convex disutility of labor ($\eta > 0$), symmetric wage adjustment costs ($\phi > 0$) and asymmetric wage adjustment costs ($\psi > 0$) where the parameter values used in each case match those estimated in Table 1.

TABLE 4—AVERAGE APPROXIMATION ACCURACY

	Price adj. costs $\xi = 37.8$	CRRA utility $\gamma = 2.52$	Convex disutility $\eta = 1.19$	Sym. wage costs $\phi = 280.4$	Asym. wage costs $\psi = 3844.4$
LQ, $n = 1$	-2.12	-2.12	-1.94	-1.94	-1.94
QC, $n = 2$	-4.31	-4.20	-3.73	-3.89	-3.26
Taylor, $n = 1$	-2.12	-2.12	-1.94	-1.94	-1.94
Taylor, $n = 2$	-4.01	-4.16	-4.08	-3.98	-3.96

Notes: Euler equation errors are reported in a decimal log scale (i.e. $\log_{10} |1 - [(c(S))^\gamma c(S^\dagger)^{-\gamma} \beta (1 + i^b(S)) \Pi(S^\dagger)^{-1}]^{-1}|$) where S denotes the current period state $S^\dagger$ the next period state. We simulate the model over 100,000 periods and calculate the mean log error in the Euler equation using a degree 11 monomial formula as per Fernández-Villaverde et al. (2015).

Across a range of parameterizations the 2nd-order approximation has a smaller average Euler equation error under both optimal policy and when a Taylor rule is used. Moving from the model with limited curvature to the full nonlinear calibration increases the average Euler equation error in the LQ approximation from 0.75% to 1.14%. By contrast the average error rate for the QC approximation increases from 0.01% to only 0.05%.

E. The role of nominal frictions

We next investigate how different nominal frictions in the New Keynesian model affect the ergodic distributions of price- and wage-inflation, the output gap, nominal interest

rates, and welfare under the two approximations. Table 5 reports the first four moments under LQ and QC, with three cases considered relative to Table 1: the approximation with quadratic price adjustment costs only ($\phi = 0$); with quadratic price and wage adjustment costs ($\psi = 0$); and additionally with asymmetric wage adjustment costs (i.e. where all nominal frictions are present).

TABLE 5—THE EFFECTS OF FRICTIONS ON THE ERGODIC MOMENTS

Mean	Price inflation	Wage inflation	Output gap	Interest rate
LQ (Quad. price adj.)	0.00	0.00	0.00	0.00
LQ (Quad. price and wage adj.)	0.00	0.00	0.00	-0.01
LQ (All frictions)	0.00	0.00	0.00	-0.01
QC (Quad. price adj.)	0.00	0.00	0.00	-0.07
QC (Quad. price and wage adj.)	0.01	0.01	0.14	0.00
QC (All frictions)	0.10	0.10	0.14	0.09
<u>Standard deviation</u>				
LQ (Quad. price adj.)	0.30	5.81	0.85	4.14
LQ (Quad. price and wage adj.)	2.24	0.21	0.25	1.89
LQ (All frictions)	2.24	0.21	0.25	1.89
QC (Quad. price adj.)	0.30	5.81	0.85	4.14
QC (Quad. price and wage adj.)	2.24	0.21	0.25	1.98
QC (All frictions)	2.24	0.35	0.35	2.12
<u>Skewness</u>				
LQ (Quad. price adj.)	0.00	0.00	0.00	0.00
LQ (Quad. price and wage adj.)	0.00	0.00	0.00	-0.01
LQ (All frictions)	0.00	0.00	0.00	-0.01
QC (Quad. price adj.)	0.07	0.00	0.00	0.01
QC (Quad. price and wage adj.)	-0.12	-0.07	0.44	0.60
QC (All frictions)	0.10	2.36	0.43	0.67
<u>Kurtosis</u>				
LQ (Quad. price adj.)	3.00	3.01	3.00	3.00
LQ (Quad. price and wage adj.)	3.01	3.02	3.01	3.00
LQ (All frictions)	3.01	3.02	3.01	3.00
QC (Quad. price adj.)	3.00	3.01	3.00	3.00
QC (Quad. price and wage adj.)	3.03	3.08	3.40	4.02
QC (All frictions)	3.06	12.20	4.80	4.70

Notes: The mean, standard deviation, skewness and kurtosis are computed using the simulated ergodic distributions under each parameterisation and policy. LQ (All frictions) and QC (All frictions) are computed under the LQ and QC approximations with all parameters set at their estimated values in Table 1. LQ (Quad. price adj.) denotes the LQ approximation with quadratic price adjustment costs only ($\phi = 0$), and LQ (Quad. price and wage adj.) denotes LQ with quadratic price and wage adjustment costs ($\psi = 0$) and with all other parameters held at their Table 1 values. The same labelling conventions apply when pre-fixed with QC.

We find that: i) there is essentially no mean shift are under any friction with LQ, while all variable means are perturbed under QC; ii) the standard deviations are quite close across approximations until asymmetry in nominal wage adjustment costs is introduced,

which *raises* the standard deviations under QC; iii) skewness is essentially zero with LQ, but positive with QC and non-zero when wage adjustment costs are symmetric (quadratic); and iv) there is strong evidence of positive excess kurtosis when compared with a normal distribution under QC, but not under LQ. Asymmetry in nominal wage adjustment costs is not the only source of asymmetry in the model. We see the symmetric components of both price and wage adjustments costs contributing to changes in all four moments as well.

Table 6 shows how the welfare results change with each friction. With any of the nominal frictions present, there are welfare losses from using a Taylor rule, and those losses are smaller under LQ than QC (column 1). However, there is also a non-monotonicity in the implied welfare losses relative to optimal policy, with the implied loss *declining* when including quadratic wage adjustment costs under QC (see column 3, row 5). Overall, we infer that while asymmetry in nominal wage adjustment costs is quantitatively important, it is not the only source of nonlinearities in the model.

TABLE 6—THE ROLE OF FRICTIONS IN WELFARE

	Taylor rule	Optimal policy	Implied welfare loss
LQ (Quad. price adj. costs)	-0.40	-0.03	0.37
LQ (Quad. price & wage adj. costs)	-0.18	-0.13	0.05
LQ (All frictions)	-0.18	-0.13	0.05
QC (Quad. price adj. costs)	-0.42	-0.03	0.39
QC (Quad. price & wage adj. costs)	-0.30	-0.13	0.17
QC (All frictions)	-0.44	-0.11	0.32

Notes: The welfare losses are calculated as per Table 2. The row labels are the same as those used in Table 5.

VI. Conclusion

We have shown how to design an n -order accurate approximation of optimal policy for a wide class of nonlinear DSGE models *analytically*. The analytical and quantitative implications of applying the approximation at $n = 1$ and $n = 2$ are then studied in the context of a New Keynesian model with costly price and wage adjustment. Allowing for a higher-order approximation opens up a new set of considerations that affect optimal policy design, and that are quantitatively important when there is curvature in the objective function and the constraints of the original nonlinear problem. We hope this approach proves useful in opening up new avenues for research on optimal policy.

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APPENDIX A: AN $(n, n + 1)$ OPTIMAL POLICY APPROXIMATION

As discussed in the main text, we use the following notation for Appendix A: a tilde accent $\sim$ denotes the deviation from steady state (e.g. $\tilde{y}_{it} := y_{it} - y_i$),³⁷ scalars have a double sub-script, e.g. y_{it} denotes the i^{th} element of the vector y_t , and if no time subscript is present denote its steady state value (e.g. y_i is the i^{th} element of the steady state vector y). Vectors only have a single time subscript (or none in steady state), and bold is reserved for stacked vectors (e.g. $\mathbf{x}_t = [y'_{t-1}, y'_t, \varepsilon'_t]'$ where $'$ is the transpose). Taylor expansions use derivatives evaluated at an approximation point (in our case the deterministic steady state) and so $\frac{\partial f_t}{\partial y_{it}|_{\mathbf{x}}}$ should be read as taking the first-partial derivative of $f(\mathbf{x}_t)$ with respect to y_{it} and then evaluating that derivative at the point $\mathbf{x}_t = \mathbf{x}$. We use the short-hand $f_t := f(\mathbf{x}_t)$ to distinguish the alternative derivative $\frac{\partial f_{t+1}}{\partial y_{it}} := \frac{\partial f(\mathbf{x}_{t+1})}{\partial y_{it}}$. Our use of multi-index notation follows the definitions presented in the main text.

We begin by deriving the objective of the $(n, n + 1)$ approximating problem. First, we compute an $n + 1$ -order Taylor approximation of the periodic Ramsey objective around the deterministic steady state $\mathbf{x}_t = \mathbf{x}$:

$$(A1) \quad f(\mathbf{x} + \tilde{\mathbf{x}}_t) = f(\mathbf{x}) + \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it}|_{\mathbf{x}}} \tilde{y}_{it} + \sum_{q=1}^p \frac{\partial f_t}{\partial \varepsilon_{qt}|_{\mathbf{x}}} \varepsilon_{qt} + \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it-1}|_{\mathbf{x}}} \tilde{y}_{it-1} \\ + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2})$$

where the multi-index $\alpha_{\mathbf{x}}$ is a $2m + p$ tuple of non-negative integers and $\mathbf{x}_t := [y'_{t-1}, y'_t, \varepsilon'_t]'$ (where y_{t-1} is the $m \times 1$ vector of lagged endogenous controls, y_t is the $m \times 1$ vector of endogenous controls, and ε_t the $p \times 1$ vector of exogenous shocks) with $\tilde{\mathbf{x}}_t := \mathbf{x}_t - \mathbf{x}$ denoting the deviation from steady state. Next consider the $n + 1$ -order approximation of the j^{th} backward-looking constraint:

$$g_j(\mathbf{x} + \tilde{\mathbf{x}}_t) = g_j(\mathbf{x}) + \sum_{i=1}^{2m+p} \frac{\partial g_{jt}}{\partial x_{it}|_{\mathbf{x}}} \tilde{x}_{it} + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2})$$

and the $n + 1$ -order approximation of the l^{th} forward-looking constraint:

$$h_l(\mathbf{x} + \tilde{\mathbf{x}}_{t+1}) = h_l(\mathbf{x}) + \sum_{i=1}^{2m+p} \frac{\partial h_{lt+1}}{\partial x_{it+1}|_{\mathbf{x}}} \tilde{x}_{it+1} + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(h_{lt+1})|_{\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2})$$

Noting the constraints are zero-valued in equilibrium, we have to an $n + 1$ -order approxi-

³⁷Stochastic shocks are the exception – they are zero valued in steady state (e.g. $\tilde{\varepsilon}_{qt} = \varepsilon_{qt}$).

mation:

$$(A2) \quad \sum_{i=1}^m \frac{\partial g_{jt}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} + \sum_{q=1}^p \frac{\partial g_{jt}}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} + \sum_{i=1}^m \frac{\partial g_{jt}}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} = - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2})$$

and

$$(A3) \quad \sum_{i=1}^m \frac{\partial h_{lt+1}}{\partial y_{it+1}} \Big|_{\mathbf{x}} \tilde{y}_{it+1} + \sum_{q=1}^p \frac{\partial h_{lt+1}}{\partial \varepsilon_{qt+1}} \Big|_{\mathbf{x}} \varepsilon_{qt+1} + \sum_{i=1}^m \frac{\partial h_{lt+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} = - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt+1})|_{\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2})$$

Also note from the steady state of the FOC of the original Ramsey FOC:

$$\left(\frac{\partial f_t}{\partial y_{it}} + \beta \frac{\partial f_{t+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} = \sum_{j=1}^J \lambda_j \left(\frac{\partial g_{jt}}{\partial y_{it}} + \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} + \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}}$$

And thus,

$$(A4) \quad \left(\frac{\partial f_t}{\partial y_{it}} + \beta \frac{\partial f_{t+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} = \sum_{j=1}^J \lambda_j \left(\frac{\partial g_{jt}}{\partial y_{it}} + \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} + \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it}$$

Using (A2), we can re-write (A4) as:

$$\begin{aligned} \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} &= - \sum_{q=1}^p \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} - \sum_{i=1}^m \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} \\ &\quad - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \sum_{i=1}^m \sum_{j=1}^J \lambda_j \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} \\ &\quad + \sum_{i=1}^m \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} - \sum_{i=1}^m \beta \frac{\partial f_{t+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} + \mathcal{O}(\epsilon^{n+2}) \end{aligned}$$

Substituting into (A1), we have:

$$\begin{aligned} f(\mathbf{x} + \tilde{\mathbf{x}}_t) &= f(\mathbf{x}) - \sum_{q=1}^p \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} - \sum_{i=1}^m \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} \\ &\quad - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \sum_{i=1}^m \sum_{j=1}^J \lambda_j \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} \\ &\quad + \sum_{i=1}^m \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} - \sum_{i=1}^m \beta \frac{\partial f_{t+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} \\ &\quad + \sum_{q=1}^p \frac{\partial f_t}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} + \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+2}) \end{aligned}$$

Collecting terms that are independent of policy (i.e. $f(\mathbf{x}) + \sum_{q=1}^p \frac{\partial f_t}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} = t.i.p$):

$$\begin{aligned}
f(\mathbf{x} + \tilde{\mathbf{x}}_t) = & - \sum_{q=1}^p \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial \varepsilon_{qt}} \Big|_{\mathbf{x}} \varepsilon_{qt} - \sum_{i=1}^m \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \\
& + \sum_{i=1}^m \sum_{j=1}^J \lambda_j \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} + \sum_{i=1}^m \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} \\
& - \sum_{i=1}^m \beta \frac{\partial f_{t+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} + \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} + \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \\
& + t.i.p + \mathcal{O}(\epsilon^{n+2})
\end{aligned}$$

Next note, when taking the conditional expectation at time $t = t_0$ of the discounted sum of the periodic objective, the terms:

$$\begin{aligned}
\text{(A5)} \quad E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(- \sum_{i=1}^m \sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} + \sum_{i=1}^m \sum_{j=1}^J \lambda_j \beta \frac{\partial g_{jt+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} \right) \right] \\
= - \sum_{i=1}^m \sum_{j=1}^J \lambda_j \frac{\partial g_{jt_0}}{\partial y_{it_0-1}} \Big|_{\mathbf{x}} \tilde{y}_{it_0-1} = t.i.p
\end{aligned}$$

$$\begin{aligned}
\text{(A6)} \quad E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(- \sum_{i=1}^m \beta \frac{\partial f_{t+1}}{\partial y_{it}} \Big|_{\mathbf{x}} \tilde{y}_{it} + \sum_{i=1}^m \frac{\partial f_t}{\partial y_{it-1}} \Big|_{\mathbf{x}} \tilde{y}_{it-1} \right) \right] \\
= \sum_{i=1}^m \frac{\partial f_{t_0}}{\partial y_{it_0-1}} \Big|_{\mathbf{x}} \tilde{y}_{it_0-1} = t.i.p
\end{aligned}$$

are both independent of policy. Further note we can use the $n+1$ -order approximation of the forward-looking constraint (A3) evaluated at time t , to show:

$$\begin{aligned}
\text{(A7)} \quad E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \sum_{i=1}^m \sum_{l=1}^L \psi_l \left(\beta^{-1} \frac{\partial h_{lt}}{\partial y_{it}} + \frac{\partial h_{lt+1}}{\partial y_{it}} \right) \Big|_{\mathbf{x}} \tilde{y}_{it} \right] \\
= -E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \sum_{l=1}^L \psi_l \beta^{-1} \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(h_{lt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + t.i.p + \mathcal{O}(\epsilon^{n+2})
\end{aligned}$$

Using (A5) to (A7), the $n+1$ -order approximation of the Ramsey objective is:

$$\begin{aligned}
E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} f(\mathbf{x} + \tilde{\mathbf{x}}_t) \right] = & -E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(\sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right] \\
& - E_{t_0} \left[\sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{l=1}^L \psi_l \beta^{-1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(h_{lt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] \\
& + E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + t.i.p + \mathcal{O}(\epsilon^{n+2})
\end{aligned}$$

This modified $n + 1$ order accurate approximation of the objective does not contain any linear terms in $\tilde{\mathbf{x}}_t$. We now define the $(n, n + 1)$ approximating problem as maximizing this modified objective, subject to an n -order approximation of the constraints. That is, the $(n, n + 1)$ approximating problem is:

$$(A8) \quad \begin{aligned} & \max_{\{y_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} W_t^{(n, n+1)} \right] \\ & s.t. : 0 = \sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+1}) \text{ for } j = 1, \dots, J \text{ and } t \geq t_0 \\ & 0 = E_t \left[\sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt+1})|_{\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + \mathcal{O}(\epsilon^{n+1}) \text{ for } l = 1, \dots, L \text{ and } t \geq t_0 - 1 \end{aligned}$$

where:

$$\begin{aligned} W_t^{(n, n+1)} := & - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} - \sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \sum_{l=1}^L \psi_l \beta^{-1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \\ & + \left[\sum_{|\alpha_{\mathbf{x}}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + t.i.p + \mathcal{O}(\epsilon^{n+2}) \end{aligned}$$

and recall $\mathbf{x}_t := [y'_{t-1}, y'_t, \varepsilon'_t]'$, where y_{t-1} is the $m \times 1$ vector of lagged endogenous controls, y_t is the $m \times 1$ vector of endogenous controls, and ε_t the $p \times 1$ vector of exogenous shocks. We now prove our main theoretical result:

PROPOSITION 1: *The FOC of the $(n, n + 1)$ approximating problem defined in (6) are equivalent to those derived from an n -order approximation of the FOC of the original Ramsey optimal policy problem defined by (3) to (5).*

PROOF:

Let $P(n)$ represent the claim made in the proposition. We proceed using a proof by induction on n , where n is a positive integer.

Verifying the claim for $n = 1$: This is the well-known Linear-Quadratic approximation – a result proven in Benigno and Woodford (2012) and is a special case of our working paper “A Quadratic-Cubic Approximation of Optimal Policy” that also proves the statement holds for $n = 2$.

Verifying $P(k) \implies P(k + 1)$ where k is an integer with $k \geq 1$. First note since the $k + 1$ -order approximations of the constraint sets of the two problems are identical, we need only show $P(k)$ implies the FOC derived from the $(k + 1, k + 2)$ approximating problem (with

respect to the vector of controls, y_t) are equivalent to those obtained from the $k + 1$ -order approximation of the Ramsey FOC up to the remainder term of the latter. Assume $P(k)$ holds. Taking the FOC of $(k + 1, k + 2)$ approximating problem (w.r.t. the i^{th} control y_{it}) we have:

$$0 = \mathcal{F}_{it}^{(k+1, k+2)} \text{ for } i = 1, \dots, m \text{ and } t \geq t_0$$

where:

$$\begin{aligned} \mathcal{F}_{it}^{(k+1, k+2)} := & \frac{\partial}{\partial y_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(f_t - \sum_{j=1}^J \lambda_j g_{jt} - \sum_{l=1}^L \psi_l \beta^{-1} h_{lt} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}} \right)}{\alpha_{\mathbf{x}}!} \\ & + \beta E_t \left[\frac{\partial}{\partial y_{it}} \sum_{|\alpha_{\mathbf{x}}|=2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(f_{t+1} - \sum_{j=1}^J \lambda_j g_{jt+1} - \sum_{l=1}^L \psi_l \beta^{-1} h_{lt+1} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}} \right]}{\alpha_{\mathbf{x}}!} \\ & - \frac{\partial}{\partial y_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\sum_{j=1}^J \tilde{\lambda}_{jt} g_{jt} + \beta^{-1} \sum_{l=1}^L \tilde{\psi}_{lt-1} h_{lt} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}} \right)}{\alpha_{\mathbf{x}}!} \\ & - E_t \left[\frac{\partial}{\partial y_{it}} \sum_{|\alpha_{\mathbf{x}}|=1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\beta \sum_{j=1}^J \tilde{\lambda}_{jt+1} g_{jt+1} + \sum_{l=1}^L \tilde{\psi}_{lt} h_{lt+1} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}} \right]}{\alpha_{\mathbf{x}}!} \end{aligned}$$

and multi-index $\alpha_{\mathbf{x}}$ is a $2m + p$ tuple of nonnegative integers and recall that $\mathbf{x}_t$ is the stacked $(2m + p) \times 1$ vector of lagged control variables (y_{t-1}) , control variables (y_t) and shocks (ε_t) and $\tilde{\mathbf{x}}_t := \mathbf{x}_t - \mathbf{x}$, and where we define normalized deviations in Lagrange multipliers from their steady state value $\tilde{\lambda}_{jt} = \lambda_{jt} - \lambda_j$ and $\tilde{\psi}_{jt} = \psi_{jt} - \psi_j$ for the backward- and forward-looking constraints respectively. Next note the $k + 1$ -order approximations of the Ramsey FOC's are given by:

$$0 = \mathcal{R}_{it}^{(k+1)} \text{ for } i = 1, \dots, m \text{ and } t \geq t_0$$

where:

(A9)

$$\begin{aligned} \mathcal{R}_{it}^{(k+1)} := & \sum_{|\alpha_{\mathbf{z}}|=0}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\frac{\partial f}{\partial y_{it}} - \sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} - \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}} }{\alpha_{\mathbf{z}}!} \\ & + E_t \left[\sum_{|\alpha_{\mathbf{z}}|=0}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \frac{\partial f_{t+1}}{\partial y_{it}} - \beta \sum_{j=1}^J \lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} - \sum_{l=1}^L \psi_{lt} \frac{\partial h_{lt+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}} \right]}{\alpha_{\mathbf{z}}!} \right] + \mathcal{O}(\epsilon^{k+2}) \end{aligned}$$

and recall $\alpha_{\mathbf{z}}$ is a separate multi-index (a $2m + p + 2L + 2J$ tuple of nonnegative integers) and that $\mathbf{z}_t$ is the stacked $(2m + p + 2J + 2L) \times 1$ vector of lagged control variables, control variables, shocks, and backward- and forward-looking Lagrange multipliers

(i.e. $\mathbf{z}_t := [\mathbf{x}'_t, \lambda'_{t-1}, \lambda'_t, \psi'_{t-1}, \psi'_t]'$ where $\lambda_t := [\lambda_{1t}, \dots, \lambda_{Jt}]'$ and $\psi_t := [\psi_{1t}, \dots, \psi_{Lt}]'$) and $\tilde{\mathbf{z}}_t := \mathbf{z}_t - \mathbf{z}$. Using the inductive hypothesis $P(k)$, which implies $\mathcal{R}_{it}^{(k)} = \mathcal{F}_{it}^{(k,k+1)}$ for $i = 1, \dots, m$ and $t \geq t_0$, we can re-write the $k+1$ -order approximation of the Ramsey FOC (A9) as:

$$\begin{aligned} 0 &= \sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\frac{\partial f}{\partial y_{it}} - \sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} - \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \\ &+ E_t \left[\sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \frac{\partial f_{t+1}}{\partial y_{it}} - \beta \sum_{j=1}^J \lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} - \sum_{l=1}^L \psi_{lt} \frac{\partial h_{lt+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] \\ &+ \mathcal{F}_{it}^{(k,k+1)} + \mathcal{O}(\epsilon^{k+2}) \text{ for } i = 1, \dots, m \text{ and } t \geq t_0. \end{aligned}$$

Next note from the FOC of the $(k+1, k+2)$ and $(k, k+1)$ approximating problems:

$$\mathcal{F}_{it}^{(k+1,k+2)} = \mathcal{F}_{it}^{(k,k+1)} + \Delta \mathcal{F}_{it}^{(k+1,k+2)} \text{ for } i = 1, \dots, m \text{ and } t \geq t_0$$

where:

$$\begin{aligned} \Delta \mathcal{F}_{it}^{(k+1,k+2)} &:= \frac{\partial}{\partial y_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(f_t - \sum_{j=1}^J \lambda_j g_{jt} - \sum_{l=1}^L \psi_l \beta^{-1} h_{lt} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \\ &+ \beta E_t \left[\frac{\partial}{\partial y_{it}} \sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(f_{t+1} - \sum_{j=1}^J \lambda_j g_{jt+1} - \sum_{l=1}^L \psi_l \beta^{-1} h_{lt+1} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] \\ &- \frac{\partial}{\partial y_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\sum_{j=1}^J \tilde{\lambda}_{jt} g_{jt} + \beta^{-1} \sum_{l=1}^L \tilde{\psi}_{lt-1} h_{lt} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \\ &- E_t \left[\frac{\partial}{\partial y_{it}} \sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\beta \sum_{j=1}^J \tilde{\lambda}_{jt+1} g_{jt+1} + \sum_{l=1}^L \tilde{\psi}_{lt} h_{lt+1} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] \end{aligned}$$

are the additional $k+1$ -order terms that are present in the $(k+1, k+2)$ solution, but not in the $(k, k+1)$ solution. Thus, if $P(k)$ is true, we need only show:

(A10)

$$\begin{aligned} \Delta \mathcal{F}_{it}^{(k+1,k+2)} &= \sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\frac{\partial f}{\partial y_{it}} - \sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} - \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \\ &+ E_t \left[\sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \frac{\partial f_{t+1}}{\partial y_{it}} - \beta \sum_{j=1}^J \lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} - \sum_{l=1}^L \psi_{lt} \frac{\partial h_{lt+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] \end{aligned}$$

for $i = 1, \dots, m$ and $t \geq t_0$ to complete the proof. We now verify (A10) holds.

First note that f is only a function of $\mathbf{x}_t$ (lagged controls, controls and shocks, and not any backward-looking or forward-looking Lagrange multipliers) so that:

$$(A11) \quad \sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\frac{\partial f_t}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} = \sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\frac{\partial f_t}{\partial y_{it}} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} = \frac{\partial}{\partial y_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right)$$

where the second equality follows from Lemma 1. Using similar arguments, it follows that:

$$(A12) \quad E_t \left[\sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \frac{\partial f_{t+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] = E_t \left[\frac{\partial}{\partial y_{it}} \beta \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_{t+1})_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right]$$

Next note by applying Lemma 2:

$$(A13) \quad \sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} = \frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \lambda_j \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \\ + \frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \tilde{\lambda}_{jt} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right)$$

$$(A14) \quad \sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\sum_{l=1}^L \beta^{-1} \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} = \beta^{-1} \frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \psi_l \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \\ + \beta^{-1} \frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \tilde{\psi}_{lt-1} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right)$$

And further that:

$$(A15) \quad E_t \left[\sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\beta \sum_{j=1}^J \lambda_{jt+1} \frac{\partial g_{jt+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] = \beta E_t \left[\frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \lambda_j \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt+1})_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \right] \\ + \beta E_t \left[\frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \tilde{\lambda}_{jt+1} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (g_{jt+1})_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \right]$$

$$(A16) \quad E_t \left[\sum_{|\alpha_{\mathbf{z}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\sum_{l=1}^L \psi_{lt} \frac{\partial h_{lt+1}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_{t+1}^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \right] = E_t \left[\frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \psi_l \left(\sum_{|\alpha_{\mathbf{x}}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt+1})_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \right] \\ + E_t \left[\frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \tilde{\psi}_{lt} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (h_{lt+1})_{|\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \right]$$

Adding (A11) and (A12), and then subtracting the sum of (A13) to (A16) verifies (A10). That is, if $P(k)$ is true, $P(k+1)$ also holds. By induction, the claim $P(n)$ holds for any integer $n \geq 1$. Lemmas 1 and 2 that were used in the proof follow.

LEMMA 1: *If $f : U \rightarrow \mathbb{R}$ is of class C^{k+2} for on an open subset $U \subset \mathbb{R}^m \times \mathbb{R}^m \times \mathbb{R}^p$, the $k+1$ -order terms in a $k+1$ Taylor expansion of the first partial derivative $\frac{\partial f}{\partial x_{it}}$ around the point $\mathbf{x}_t = \mathbf{x}$, is equal to the partial derivative of the $k+2$ order terms of the $k+2$ Taylor expansion of the original function f at $\mathbf{x}_t = \mathbf{x}$. That is,*

$$\sum_{|\alpha_{\mathbf{x}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\frac{\partial f_t}{\partial x_{it}} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} = \frac{\partial}{\partial x_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right)$$

PROOF:

This is straightforward to show by expanding out the multi-indices, evaluating the derivative, noting that if f is of class C^{k+2} the order of differentiation can be interchanged (Clairaut-Schwarz's theorem), and then using a change of index. Specifically,

$$\begin{aligned} & \frac{\partial}{\partial x_{it}} \left(\sum_{|\alpha_{\mathbf{x}}|=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \\ &= \frac{\partial \sum_{\alpha_1+\dots+\alpha_{2m+p}=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)_{|\mathbf{x}} \prod_{r=1}^{2m+p} \tilde{x}_{rt}^{\alpha_r}}{\prod_{r=1}^{2m+p} \alpha_r!}}{\partial x_{it}} \\ &= \sum_{\alpha_1+\dots+\alpha_i+\dots+\alpha_{2m+p}=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} (f_t)_{|\mathbf{x}} \prod_{r \neq i} \tilde{x}_{rt}^{\alpha_r} \alpha_i \tilde{x}_{it}^{\alpha_i-1}}{\prod_{r=1}^{2m+p} \alpha_r!} \\ &= \sum_{\alpha_1+\dots+(\alpha_i-1)+\dots+\alpha_{2m+p}=k+1} \frac{(\partial_{x_{1t}}^{\alpha_1} \dots \partial_{x_{2m+p,t}}^{\alpha_{2m+p}} f_t)_{|\mathbf{x}} \prod_{r \neq i} \tilde{x}_{rt}^{\alpha_r} \tilde{x}_{it}^{\alpha_i-1}}{\prod_{r \neq i} \alpha_r! (\alpha_i-1)!} \\ &= \sum_{\alpha_1+\dots+(\alpha_i-1)+\dots+\alpha_n=k+1} \frac{(\partial_{x_{1t}}^{\alpha_1} \dots \partial_{x_{it}}^{\alpha_i-1} \dots \partial_{x_{2m+p,t}}^{\alpha_{2m+p}} \partial_{x_{it}} f_t)_{|\mathbf{x}} \prod_{r \neq i} \tilde{x}_{rt}^{\alpha_r} \tilde{x}_{it}^{\alpha_i-1}}{\prod_{r \neq i} \alpha_r! (\alpha_i-1)!} \\ &= \sum_{\alpha_1+\dots+\alpha_s+\dots+\alpha_n=k+1} \frac{(\partial_{x_{1t}}^{\alpha_1} \dots \partial_{x_{st}}^{\alpha_s} \partial_{x_{2m+p}}^{\alpha_{2m+p}} \partial_{x_{it}} f_t)_{|\mathbf{x}} \prod_{r \neq i} \tilde{x}_{rt}^{\alpha_r} \tilde{x}_{it}^s}{\prod_{r \neq i} \alpha_r! \alpha_s!} \\ &= \sum_{|\alpha_{\mathbf{x}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}} \left(\frac{\partial f_t}{\partial x_{it}} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \end{aligned}$$

as required.

LEMMA 2: If the functions $g_j : U \rightarrow \mathbb{R}$ are of class C^{k+2} on an open subset $U \subset \mathbb{R}^m \times \mathbb{R}^m \times \mathbb{R}^p$ for $j = 1, \dots, J$, the sum of the $k+1$ -order terms in the $k+1$ -order expansion of the sum of the Ramsey Lagrange multipliers weighted by the partial derivatives of the backward-looking constraints with respect to the i^{th} control (y_{it}) are equal to the partial derivative ($\frac{\partial}{\partial y_{it}}$) of the sum of:

- a) the steady state Lagrange multipliers multiplied by the $k+2$ -order expansion of the backward-looking constraints $\left(\sum_{j=1}^J \lambda_j \left(\sum_{|\alpha_{\mathbf{x}}|=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right)$; and
- b) the steady state deviations in the Lagrange multipliers multiplied by the $k+1$ -order expansion of the backward-looking constraints $\sum_{j=1}^J \tilde{\lambda}_{jt} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right)$. That is,

$$\begin{aligned} \sum_{|\alpha_{\mathbf{z}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} \right) |_{\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} &= \frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \lambda_j \left(\sum_{|\alpha_{\mathbf{x}}|=k+2} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \\ &+ \frac{\partial}{\partial y_{it}} \left(\sum_{j=1}^J \tilde{\lambda}_{jt} \left(\sum_{|\alpha_{\mathbf{x}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right) \right) \end{aligned}$$

An analogous results holds for the forward-looking constraints.

PROOF:

First note that since the function $\sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}}$ is only a function of $\mathbf{x}_t$ and $\lambda_t = [\lambda_{1t}, \dots, \lambda_{Jt}]'$, and not the forward-looking multipliers, we can evaluate the $k+1$ -order terms in the expansion as:

$$\begin{aligned} \text{(A17)} \quad & \sum_{|\alpha_{\mathbf{z}}|=k+1} \frac{\mathbf{D}^{\alpha_{\mathbf{z}}} \left(\sum_{j=1}^J \lambda_{jt} \frac{\partial g_{jt}}{\partial y_{it}} \right) |_{\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_{\mathbf{z}}}}{\alpha_{\mathbf{z}}!} \\ &= \sum_{|\alpha_{\lambda}|=k+1} \frac{\left(\partial_{x_1}^{\alpha_1} \dots \partial_{x_{2m+p}}^{\alpha_{2m+p}} \partial_{\lambda_1}^{\alpha_{2m+p+1}} \dots \partial_{\lambda_l}^{\alpha_{2m+p+J}} \left(\sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it}} \right) \right) |_{\mathbf{x}, \lambda} \tilde{x}_{1t}^{\alpha_1} \dots \tilde{\lambda}_{Jt}^{\alpha_{2m+p+J}}}{\alpha_{\lambda}!} \end{aligned}$$

where α_{λ} is $(2m+p+J)$ tuple allowing one to index over the lagged controls (y_{t-1}), time- t controls (y_t), the shocks (ε_t) and the backward-looking Lagrange multipliers (λ_{jt}). Next note that only the zero-order and 1st-order terms in the backward-looking Lagrange multipliers are non-zero in the $k+1$ -order terms in the $k+1$ -order expansion of $\sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it}}$.

Thus, we can evaluate (A17) as:

$$\begin{aligned}
& \sum_{|\alpha_\lambda|=k+1}^{k+1} \frac{\left(\partial_{x_1}^{\alpha_1} \dots \partial_{x_{2m+p}}^{\alpha_{2m+p}} \partial_{\lambda_1}^{\alpha_{2m+p+1}} \dots \partial_{\lambda_l}^{\alpha_{2m+p+J}} \left(\sum_{j=1}^J \lambda_j \frac{\partial g_{jt}}{\partial y_{it}} \right) \right)_{|\mathbf{x}, \lambda}}{\alpha_\lambda!} (\tilde{x}_{1t})^{\alpha_1} \dots (\tilde{\lambda}_{Jt})^{\alpha_{2m+p+J}} \\
&= \sum_{|\alpha_\mathbf{x}|=k}^k \sum_{j=1}^J \frac{\left(\partial_{x_1}^{\alpha_1} \dots \partial_{x_{2m+p}}^{\alpha_{2m+p}} \frac{\partial g_{jt}}{\partial y_{it}} \right)_{|\mathbf{x}}}{\prod_{i=1}^{2m+p} (\alpha_{\mathbf{x},i})! 1!} \tilde{x}_{1t}^{\alpha_1} \dots \tilde{x}_{2m+p,t}^{\alpha_{2m+p}} \tilde{\lambda}_{jt} \\
&+ \sum_{|\alpha_\mathbf{x}|=k+1}^{k+1} \sum_{j=1}^J \frac{\left(\partial_{x_1}^{\alpha_1} \dots \partial_{x_{2m+p}}^{\alpha_{2m+p}} \frac{\partial g_{jt}}{\partial y_{it}} \right)_{|\mathbf{x}}}{\prod_{i=1}^{2m+p} (\alpha_{\mathbf{x},i})! 0!} \tilde{x}_{1t}^{\alpha_1} \dots \tilde{x}_{2m+p,t}^{\alpha_{2m+p}} \lambda_j \\
&= \sum_{j=1}^J \tilde{\lambda}_{jt} \left(\sum_{|\alpha_\mathbf{x}|=k}^k \frac{\mathbf{D}^{\alpha_\mathbf{x}} \left(\frac{\partial g_{jt}}{\partial y_{it}} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} \right) + \sum_{j=1}^J \lambda_j \left(\sum_{|\alpha_\mathbf{x}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_\mathbf{x}} \left(\frac{\partial g_{jt}}{\partial y_{it}} \right)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} \right)
\end{aligned}$$

Finally noting that $\tilde{\lambda}_{jt} = \lambda_{jt} - \lambda_j$ and applying Lemma 1 proves the required result. It is straightforward to apply the same arguments and prove an analogous result holds for the forward-looking multipliers as well:

$$\begin{aligned}
& \sum_{|\alpha_\mathbf{z}|=k+1}^{k+1} \frac{\left(\mathbf{D}^{\alpha_\mathbf{z}} \beta^{-1} \sum_{l=1}^L \psi_{lt-1} \frac{\partial h_{lt}}{\partial y_{it}} \right)_{|\mathbf{z}} \tilde{\mathbf{z}}_t^{\alpha_\mathbf{z}}}{\alpha_\mathbf{z}!} = \beta^{-1} \frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \psi_l \left(\sum_{|\alpha_\mathbf{x}|=k+2}^{k+2} \frac{\mathbf{D}^{\alpha_\mathbf{x}} (h_{lt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} \right) \right) \\
&+ \beta^{-1} \frac{\partial}{\partial y_{it}} \left(\sum_{l=1}^L \tilde{\psi}_{lt-1} \left(\sum_{|\alpha_\mathbf{x}|=k+1}^{k+1} \frac{\mathbf{D}^{\alpha_\mathbf{x}} (h_{lt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} \right) \right)
\end{aligned}$$

A1. Sufficiency Conditions

Here, we briefly discuss sufficiency conditions drawing on earlier work by Benigno and Woodford (2012). Recall the per-period $(n, n+1)$ objective is:

$$\begin{aligned}
W_t^{(n,n+1)}(\tilde{y}_{t-1}, \tilde{y}_t, \varepsilon_t) &:= - \sum_{|\alpha_\mathbf{x}|=2}^{n+1} \sum_{j=1}^J \lambda_j \frac{\mathbf{D}^{\alpha_\mathbf{x}} (g_{jt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} - \sum_{|\alpha_\mathbf{x}|=2}^{n+1} \sum_{l=1}^L \psi_l \beta^{-1} \frac{\mathbf{D}^{\alpha_\mathbf{x}} (h_{lt})_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} \\
&+ \sum_{|\alpha_\mathbf{x}|=2}^{n+1} \frac{\mathbf{D}^{\alpha_\mathbf{x}} (f_t)_{|\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_\mathbf{x}}}{\alpha_\mathbf{x}!} + t.i.p + \mathcal{O}(\epsilon^{n+2})
\end{aligned}$$

where $\tilde{\mathbf{x}}_t := \mathbf{x}_t - \mathbf{x}$, $\mathbf{x}_t := [y'_{t-1}, y'_t, \varepsilon'_t]'$, $\tilde{y}_{t-1} = y_{t-1} - y$, $\tilde{y}_t = y_t - y$ and $\tilde{\varepsilon}_t = \varepsilon_t$. Let $\mathcal{H}$ be a real-valued Hilbert space and $\mathcal{H}_1 \subset \mathcal{H}$ the sub-space of endogenous stochastic processes $\{\tilde{y}_t\}_{t \geq t_0}$ for given exogenous shock process $\{\varepsilon_t\}_{t \geq t_0}$ such that:

$$\mathcal{W}_{t_0}^{(n,n+1)} := E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} W_t^{(n,n+1)}(\tilde{y}_{t-1}, \tilde{y}_t, \varepsilon_t) < \infty$$

and taking $\tilde{y}_{t_0-1}$ as given. For $\{\tilde{y}_t\}_{t \geq t_0}$ to maximize $\mathcal{W}_{t_0}^{(n,n+1)}$ subject to:

$$(A18) \quad 0 = \sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(g_{jt})|_{\mathbf{x}} \tilde{\mathbf{x}}_t^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} + \mathcal{O}(\epsilon^{n+1}) \text{ for } j = 1, \dots, J \text{ and } \forall t \geq t_0$$

$$(A19) \quad 0 = E_t \left[\sum_{|\alpha_{\mathbf{x}}|=1}^n \frac{\mathbf{D}^{\alpha_{\mathbf{x}}}(h_{lt+1})|_{\mathbf{x}} \tilde{\mathbf{x}}_{t+1}^{\alpha_{\mathbf{x}}}}{\alpha_{\mathbf{x}}!} \right] + \mathcal{O}(\epsilon^{n+1}) \text{ for } l = 1, \dots, L \text{ and } \forall t \geq t_0 - 1$$

it is necessary and sufficient that (i) there exist Lagrange multiplier processes $\{\lambda_{jt}\}_{t \geq t_0}$ and $\{\psi_{lt}\}_{t \geq t_0-1}$, at least some non-zero for $j = 1, \dots, J$ and $l = 1, \dots, L$, such that the FOC of the $(n, n+1)$ approximating problem are satisfied for all $t \geq t_0$ and (ii) for any real-valued stochastic process $\{\tilde{y}_t\}_{t \geq t_0}$ that also lies in the sub-space $\mathcal{H}_1 \subset \mathcal{H}$ that additionally satisfies the constraints (A18) to (A19), it must be the case that:

$$E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} W_t^{(n, n+1)}(\tilde{y}_{t-1}, \tilde{y}_t, \varepsilon_t) \leq E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} W_t^{(n, n+1)}(\tilde{y}_{t-1}, \tilde{y}_t, \varepsilon_t)$$

taking $\tilde{y}_{t_0-1} = \tilde{y}_{t_0-1}$ as given, and where if the inequality is strict then the maximizing sequence $\{\tilde{y}_t\}_{t \geq t_0}$ is unique.

APPENDIX B: QC OPTIMAL POLICY IN THE NEW KEYNESIAN MODEL WITH COSTLY PRICE
AND WAGE ADJUSTMENT

B1. Optimal Ramsey Policy

The Ramsey policy problem is to maximize household utility choosing sequences of consumption, labor, output, inflation, wage inflation, and real wages to maximize:

$$(B1) \quad \max_{\{C_t, H_t, Y_t, \Pi_t, \Pi_t^w, \frac{W_t}{P_t}\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} (U(C_t; Q_t) - V(H_t)) \right]$$

subject to the aggregate resource constraint:

$$(B2) \quad (1 - \Phi_t^\pi) Y_t - \frac{W_t}{P_t} H_t \Phi_t = C_t$$

the nonlinear NKWPC:

$$(B3) \quad \frac{1 - \Phi_t}{\theta_t - 1} + \frac{\partial \Phi_t}{\partial \Pi_t^w} \Pi_t^w = \frac{V_H(H_t)}{U_c(C_t; Q_t)} \frac{\theta_t}{\theta_t - 1} \left(\frac{W_t}{P_t} \right)^{-1} \frac{1}{1 - \tau^w} + \beta E_t \left(\frac{U_c(C_{t+1}; Q_{t+1})}{U_c(C_t; Q_t)} \Pi_{t+1}^{-1} \frac{H_{t+1}}{H_t} \frac{\partial \Phi_{t+1}}{\partial \Pi_{t+1}^w} (\Pi_{t+1}^w)^2 \right)$$

the nonlinear NKPC:

$$(B4) \quad \frac{1 - \Phi_t^\pi}{\varepsilon - 1} + \Pi_t \frac{\partial \Phi_t^\pi}{\partial \Pi_t} = \frac{\varepsilon(1 - \varsigma)}{\varepsilon - 1} \frac{W_t}{P_t} \frac{\Upsilon_t}{A_t} + \beta E_t \left[\frac{U_c(C_{t+1}; Q_{t+1}) Y_{t+1}}{U_c(C_t; Q_t) Y_t} \frac{\partial \Phi_{t+1}^\pi}{\partial \Pi_{t+1}} \Pi_{t+1} \right]$$

and the production function:

$$(B5) \quad Y_t = A_t H_t$$

and given the definition of wage inflation $\Pi_t^w := \frac{W_t/P_t}{W_{t-1}/P_{t-1}} \Pi_t$, and taking exogenous shock sequences $\{A_t, Q_t, \Upsilon_t, \theta_t\}_{t \geq t_0}$ as given.

Note that in the subsequent LQ and QC approximations of this problem we will assume $\tau^w = 1 - \theta$ and $\varsigma = 1 - \varepsilon^{-1}$ so that the steady state distortions associated with monopolistic competition in goods and labor markets are removed and do not affect optimal policy.³⁸

B2. A QC Approximation

We first derive optimal policy under QC in the model with all shocks (demand, productivity, cost-push and wage-markup), allowing for non-unitary household consumption risk aversion and price and wage adjustment costs. We then remark on the corresponding LQ approximation. Qualitative results for the simpler economy discussed in the main text are discussed in Appendix D.

³⁸In effect, this eliminates any incentive for policymakers to try and offset steady state inefficiencies through monetary policy.

PROPOSITION 3: *The QC approximation of the optimal Ramsey problem around the efficient steady state is:*

$$\min_{\{\hat{i}_t^b, x_t, \pi_t, \pi_t^w, w_t^g\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(L_t^{QC} \right) \right]$$

$$\begin{aligned} L_t^{QC} := & (\gamma + \eta) \left(1 + \omega_t^{QC} \right) \frac{x_t^2}{2} + \xi \left(1 + \omega_t^{QC} \right) \frac{\pi_t^2}{2} - \left((\gamma - 1)^2 - (1 + \eta)^2 \right) \frac{1}{3!} x_t^3 \\ & - \xi (\gamma - 1) \frac{\pi_t^2 x_t}{2} + \xi \frac{\pi_t^3}{2} + \frac{1}{2} \phi \left(\left(1 + \omega_t^{QC} \right) \pi_t^{w^2} + \left(1 - \frac{1}{3} \psi \right) \pi_t^{w^3} + \pi_t^{w^2} w_t^g - (\gamma - 1) \pi_t^{w^2} x_t \right) \end{aligned}$$

with state-contingent weight $\omega_t^{QC} := \frac{1+\eta}{\gamma+\eta} (q_t - (\gamma - 1) a_t)$, and where the minimization is subject to the 2nd-order expansion of the NKWPC:

$$\pi_t^w + (3 - \psi) \frac{\pi_t^{w^2}}{2} = \frac{1}{\phi(\theta - 1)} (J_t) + \beta E_t \left[\pi_{t+1}^w + (3 - \psi) \frac{\pi_{t+1}^{w^2}}{2} \right] + \beta E_t [\Xi_{t+1}^w \pi_{t+1}^w]$$

the marginal disutility of supplying labor:

$$\begin{aligned} J_t := & (\gamma + \eta) x_t - w_t^g + \frac{1}{2} ((\gamma + \eta) x_t - w_t^g)^2 + \left(\frac{1}{\theta - 1} \right) \hat{\theta}_t (w_t^g - (\gamma + \eta) x_t) \\ & - \gamma \xi \frac{\pi_t^2}{2} - (\gamma - 1) \phi \frac{\pi_t^{w^2}}{2} + \hat{\theta}_t - \frac{1 + \theta}{\theta - 1} \frac{\hat{\theta}_t^2}{2} \end{aligned}$$

the 2nd-order expansions of the NKPC and NKIS curves:

$$\begin{aligned} \pi_t + \frac{3}{2} \pi_t^2 &= \kappa s_t^g + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right] + \beta E_t [\Xi_{t+1}^\pi \pi_{t+1}] \\ E_t [r_{t+1}^g] &= E_t \left[\gamma \Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g - \gamma \Delta x_{t+1})^2 - \gamma \frac{\xi}{2} \Delta \pi_{t+1}^2 - \gamma \frac{\phi}{2} \Delta \pi_{t+1}^{w^2} \right] \end{aligned}$$

with real marginal cost, interest-rate, wage and output gaps and wage inflation given by:

$$\begin{aligned} s_t^g &:= w_t^g + \mu_t + \frac{1}{2} (w_t^g + \mu_t)^2 - \xi \frac{\pi_t^2}{2}, & r_{t+1}^g &:= \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn} \\ w_t^g &:= w_t - w_t^n, & x_t &:= y_t - y_t^n, & \pi_t^w &:= \Delta w_t + \pi_t \end{aligned}$$

the flexible price and wage equilibrium is:

$$y_t^n := \frac{q_t}{\gamma + \eta} + \frac{1 + \eta}{\gamma + \eta} a_t, \quad \hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}, \quad w_t^n := a_t$$

and given covariance terms:

$$\begin{aligned} \Xi_{t+1}^w &:= \Delta y_{t+1}^n - \Delta w_{t+1}^n - (\gamma - 1) \Delta x_{t+1} - \left(\hat{r}_{t+1}^{bn} + \pi_{t+1} - \pi_{t+1}^w \right) \\ \Xi_{t+1}^\pi &:= \Delta y_{t+1}^n - \hat{r}_{t+1}^{bn} - (\gamma - 1) \Delta x_{t+1} \end{aligned}$$

While the QC approximation includes all terms, the LQ approximation includes only quadratic terms in the objective and linear terms in the constraints.

PROOF:

Our strategy is as follows. First, take 3rd-order expansions of household felicity and the aggregate resource constraint and use them to derive a 3rd-order expansion of the objective that is free of endogenous linear terms. Next re-formulate this objective in terms of deviations from the frictionless equilibrium. Then, take 2nd-order expansions of constraints and re-write them in terms of deviations from the frictionless equilibrium. Finally, apply Proposition 1. Further detail on each step follows.

B.2.1. A 3RD-ORDER EXPANSION OF HOUSEHOLD FELICITY AND THE RESOURCE CONSTRAINT

The felicity function for households is $\mathcal{W}_t := U(C_t; Q_t) - V(H_t)$ where $U(C_t; Q_t) := Q_t C_t^{1-\gamma} / (1-\gamma)$ and $V(H_t) := H_t^{1+\eta} / (1+\eta)$. Taking a 3rd-order expansion of $U(C_t; Q_t)$:

$$U(C_t; Q_t) = \overline{C\overline{U}}_c \left(c_t + \frac{1}{2} (1-\gamma) c_t^2 + (\gamma-1)^2 \frac{1}{3!} c_t^3 + \frac{1}{2} q_t^2 c_t + \frac{1}{2} (1-\gamma) q_t c_t^2 + q_t c_t \right) + t.i.p + \mathcal{O}(\epsilon^4)$$

A 3rd-order expansion of the disutility from labor gives:

$$V(H_t) = V(\overline{H}) + \overline{H\overline{V}}_H \left(h_t + \frac{1}{2!} (1+\eta) h_t^2 + \frac{1}{3!} ((\eta+1)^2) h_t^3 \right) + \mathcal{O}(\epsilon^4)$$

Combining these results, the 3rd-order expansion of household felicity is:

$$\begin{aligned} \mathcal{W}_t = & \overline{C\overline{U}}_c \left(c_t + \frac{1}{2} (1-\gamma) c_t^2 + (\gamma-1)^2 \frac{1}{3!} c_t^3 + \frac{1}{2} q_t^2 c_t + \frac{1}{2} (1-\gamma) q_t c_t^2 + q_t c_t \right) \\ & - \overline{H\overline{V}}_H' \left(h_t + \frac{1}{2!} (1+\eta) h_t^2 + \frac{1}{3!} ((\eta+1)^2) h_t^3 \right) + t.i.p + \mathcal{O}(\epsilon^4) \end{aligned}$$

Note from the steady state intra-temporal household optimality condition, $U_c(\overline{Y}) \overline{Y} = V_H(\overline{H}) \overline{H}$, and the steady state aggregate resource constraint implies $\overline{Y} = \overline{C}$ so we have:

$$\mathcal{W}_t = \overline{Y\overline{U}}_c \left(c_t + \frac{1}{2} (1-\gamma) c_t^2 + (\gamma-1)^2 \frac{1}{3!} c_t^3 + \frac{1}{2} q_t^2 c_t + \frac{1}{2} (1-\gamma) q_t c_t^2 + q_t c_t \right) - \left(h_t + \frac{1}{2!} (1+\eta) h_t^2 + \frac{1}{3!} ((\eta+1)^2) h_t^3 \right) + t.i.p + \mathcal{O}(\epsilon^4)$$

Next note the aggregate resource constraint is given by:

$$(B6) \quad C_t = Y_t - \frac{\xi}{2} (\Pi_t - 1)^2 Y_t - \frac{W_t}{P_t} H_t \Phi_t$$

where $\Phi_t := \phi \left(\frac{\exp(-\psi(\Pi_t^w - 1)) + \psi(\Pi_t^w - 1) - 1}{\psi^2} \right)$. Taking a 3rd-order expansion we have:

$$\begin{aligned} c_t = & y_t + \frac{y_t^2}{2} + \frac{1}{3!} y_t^3 - \frac{\xi}{2} \pi_t^2 - \frac{\xi}{2} \pi_t^3 - \frac{\xi}{2} \pi_t^2 y_t - \frac{c_t^2}{2} - \frac{c_t^3}{3!} \\ & - \frac{1}{2} \phi \left(\pi_t^{w^2} + \left(1 - \frac{1}{3} \psi \right) \pi_t^{w^3} + \pi_t^{w^2} w_t + \pi_t^{w^2} h_t \right) + \mathcal{O}(\epsilon^4) \end{aligned}$$

Substituting for consumption terms using the 3rd-order approximation of the aggregate resource constraint:

$$\mathcal{W}_t = \overline{YU}_c \left(\begin{aligned} & y_t + \frac{y_t^2}{2} + \frac{1}{3!} y_t^3 - \xi \frac{\pi_t^2}{2} - \xi \frac{\pi_t^3}{2} - \xi \frac{\pi_t^2 y_t}{2} - \frac{1}{2} \phi \left(\pi_t^{w^2} + \left(1 - \frac{1}{3} \psi\right) \pi_t^{w^3} + \pi_t^{w^2} w_t + \pi_t^{w^2} h_t \right) \\ & - \frac{c_t^2}{2} - \frac{c_t^3}{3!} + \frac{1}{2} (1 - \gamma) c_t^2 + (\gamma - 1)^2 \frac{1}{3!} c_t^3 + \frac{1}{2} q_t^2 c_t + \frac{1}{2} (1 - \gamma) q_t c_t^2 + q_t c_t \\ & - \left(h_t + \frac{1}{2!} (1 + \eta) h_t^2 + \frac{1}{3!} \left((\eta + 1)^2 \right) h_t^3 \right) \end{aligned} \right) + t.i.p + \mathcal{O}(\epsilon^4)$$

Using the 2nd- and 1st-order expansions of the aggregate resource constraint (B6), we can substitute for 2nd- and 3rd-order terms in welfare. Doing so and omitting terms that are independent of policy and the approximation error for brevity:

$$\mathcal{W}_t = \overline{YU}_c \left(\begin{aligned} & y_t + \frac{y_t^2}{2} - \xi \frac{\pi_t^2}{2} - \xi \frac{\pi_t^3}{2} - \xi \frac{\pi_t^2 y_t}{2} - \frac{1}{2} \phi \left(\pi_t^{w^2} + \left(1 - \frac{1}{3} \psi\right) \pi_t^{w^3} + \pi_t^{w^2} w_t + \pi_t^{w^2} h_t \right) \\ & - \frac{1}{2} \gamma \left(-\phi y_t \pi_t^{w^2} - \xi \pi_t^2 y_t + y_t^2 \right) + \left((\gamma - 1)^2 \right) \frac{1}{3!} y_t^3 + \frac{1}{2} q_t^2 y_t + \frac{1}{2} (1 - \gamma) q_t y_t^2 \\ & + q_t \left(y_t - \xi \frac{\pi_t^2}{2} - \phi \frac{\pi_t^{w^2}}{2} \right) - \left(h_t + \frac{1}{2!} (1 + \eta) h_t^2 + \frac{1}{3!} \left((\eta + 1)^2 \right) h_t^3 \right) \end{aligned} \right)$$

Using the production technology $h_t = y_t - a_t$ and re-writing:

$$\mathcal{W}_t = \overline{YU}_c \left(\begin{aligned} & -(\gamma + \eta) \frac{y_t^2}{2} - \xi \frac{\pi_t^2}{2} - \xi \frac{\pi_t^3}{2} - \xi \frac{\pi_t^2 y_t}{2} - \frac{1}{2} \phi \left(\pi_t^{w^2} + \left(1 - \frac{1}{3} \psi\right) \pi_t^{w^3} + \pi_t^{w^2} w_t + \pi_t^{w^2} h_t \right) \\ & - \frac{1}{2} \gamma \left(-\phi y_t \pi_t^{w^2} - \xi \pi_t^2 y_t \right) + \left((\gamma - 1)^2 \right) \frac{1}{3!} y_t^3 + \frac{1}{2} q_t^2 y_t + \frac{1}{2} (1 - \gamma) q_t y_t^2 \\ & - z_t \left(\xi \frac{\pi_t^2}{2} + \phi \frac{\pi_t^{w^2}}{2} \right) + (q_t + (1 + \eta) a_t) y_t - \frac{1}{3!} \left((\eta + 1)^2 \right) (y_t - a_t)^3 \end{aligned} \right)$$

B.2.2. REWRITE THE OBJECTIVE

Next we can use the natural values for output and wages (i.e. their frictionless or flexible-wage-and-price equilibrium values) $y_t^n = \frac{q_t}{\gamma + \eta} + \frac{1 + \eta}{\gamma + \eta} a_t$ and $w_t^n = a_t$ (where the latter follows from the fact that the log deviation in real marginal costs are zero in the frictionless equilibrium). Using this and again dropping terms that are independent of policy, we can write:

$$\mathcal{W}_t = \overline{YU}_c \left(\begin{aligned} & -(\gamma + \eta) \frac{(y_t - y_t^n)^2}{2} - \xi \frac{\pi_t^2}{2} - \xi \frac{\pi_t^3}{2} - \xi \frac{\pi_t^2 y_t}{2} - \frac{1}{2} \phi \left(\pi_t^{w^2} + \left(1 - \frac{1}{3} \psi\right) \pi_t^{w^3} + \pi_t^{w^2} w_t + \pi_t^{w^2} h_t \right) \\ & - \frac{1}{2} \gamma \left(-\phi y_t \pi_t^{w^2} - \xi \pi_t^2 y_t \right) + \left((\gamma - 1)^2 - (\eta + 1)^2 \right) \frac{1}{3!} (y_t - y_t^n)^3 - q_t \left(\xi \frac{\pi_t^2}{2} + \phi \frac{\pi_t^{w^2}}{2} \right) \\ & - (\gamma - 1) (\gamma + \eta) y_t^n y_t + \frac{1}{2} (\gamma - 1) (\gamma + \eta) y_t^n y_t^2 - \frac{1}{2} (\gamma + \eta) q_t y_t^2 + (\gamma + \eta) y_t^n q_t y_t \end{aligned} \right)$$

With some algebra, collecting terms and substituting for labor, we have:

$$\mathcal{W}_t = \overline{YU}_c \left(\begin{aligned} & -(\gamma + \eta) \left(1 + \omega_t^{QC} \right) \frac{x_t^2}{2} - \xi \left(1 + \omega_t^{QC} \right) \frac{\pi_t^2}{2} - \xi \frac{\pi_t^3}{2} + \xi (\gamma - 1) \frac{\pi_t^2 x_t}{2} \\ & - \frac{1}{2} \phi \left(\left(1 + \omega_t^{QC} \right) \pi_t^{w^2} + \left(1 - \frac{1}{3} \psi \right) \pi_t^{w^3} + \pi_t^{w^2} w_t^g - (\gamma - 1) \pi_t^{w^2} x_t \right) \\ & + \left((\gamma - 1)^2 - (\eta + 1)^2 \right) \frac{1}{3!} x_t^3 \end{aligned} \right)$$

where $\omega_t^{QC} := \left(1 - \frac{(\gamma-1)}{\gamma+\eta}\right) q_t - (\gamma-1) \frac{1+\eta}{\gamma+\eta} a_t$.

B.2.3. 2ND-ORDER EXPANSIONS OF THE CONSTRAINTS

Recall the non-linear New Keynesian Wage Phillips Curve (NKWPC) is given by:

$$\frac{1 - \Phi_t}{\theta_t - 1} + \frac{\partial \Phi_t}{\partial \Pi_t^w} \Pi_t^w = \frac{V_H(H_t)}{U_c(C_t; Q_t)} \frac{\theta_t}{\theta_t - 1} \frac{1}{1 - \tau^w} \left(\frac{W_t}{P_t} \right)^{-1} + \beta E_t \left[\frac{U_c(C_{t+1}; Q_{t+1})}{U_c(C_t; Q_t)} \Pi_{t+1}^{-1} \frac{H_{t+1}}{H_t} \frac{\partial \Phi_{t+1}}{\partial \Pi_{t+1}^w} (\Pi_{t+1}^w)^2 \right]$$

Taking a 2nd-order expansion and with some algebra it is straightforward to verify

$$\begin{aligned} \phi(\pi_t^w) + \phi\left(3 - \psi - \frac{1}{\theta - 1}\right) \frac{\pi_t^{w^2}}{2} &= \frac{1}{\theta - 1} \left(\eta h_t + \eta^2 \frac{h_t^2}{2} + \gamma c_t + \gamma^2 \frac{c_t^2}{2} - w_t + \frac{w_t^2}{2} - q_t + \frac{q_t^2}{2} \right) \\ &\quad + \frac{1}{\theta - 1} (\eta \gamma h_t c_t - \eta h_t w_t - \gamma c_t w_t - \eta h_t q_t - \gamma c_t q_t + q_t w_t) \\ &\quad + \frac{1}{\theta - 1} \left(\hat{\theta}_t - \frac{1 + \theta}{\theta - 1} \frac{\hat{\theta}_t^2}{2} \right) - \left(\frac{1}{\theta - 1} \right)^2 \left(\gamma \hat{\theta}_t c_t - \hat{\theta}_t q_t + \eta \hat{\theta}_t h_t - \hat{\theta}_t w_t \right) \\ &\quad + \beta \phi E_t \left[\begin{aligned} &\pi_{t+1}^w - (\psi - 5) \frac{\pi_{t+1}^{w^2}}{2} - \gamma c_{t+1} \pi_{t+1}^w + \gamma c_t \pi_{t+1}^w \\ &+ q_{t+1} \pi_{t+1}^w - q_t \pi_{t+1}^w + h_{t+1} \pi_{t+1}^w - h_t \pi_{t+1}^w - \pi_{t+1} \pi_{t+1}^w \end{aligned} \right] \\ &\quad + \mathcal{O}(\epsilon^3) \end{aligned}$$

Substituting in for consumption terms using a 2nd-order expansion of the aggregate resource constraint and using the production function, we have:

$$\begin{aligned} \phi(\pi_t^w) + \phi(3 - \psi) \frac{\pi_t^{w^2}}{2} &= \frac{1}{\theta - 1} (J_t) + \beta \phi E_t \left[\pi_{t+1}^w + (5 - \psi) \frac{\pi_{t+1}^{w^2}}{2} \right] \\ &\quad + \beta \phi E_t \left[\begin{aligned} &-\gamma y_{t+1} \pi_{t+1}^w + \gamma y_t \pi_{t+1}^w + q_{t+1} \pi_{t+1}^w \\ &- q_t \pi_{t+1}^w + h_{t+1} \pi_{t+1}^w - h_t \pi_{t+1}^w - \pi_{t+1} \pi_{t+1}^w \end{aligned} \right] + \mathcal{O}(\epsilon^3) \end{aligned}$$

where:

$$\begin{aligned} J_t &:= \eta(y_t - a_t) + \eta^2 \frac{(y_t - a_t)^2}{2} + \gamma y_t + \gamma^2 \frac{y_t^2}{2} - w_t + \frac{w_t^2}{2} - q_t + \frac{q_t^2}{2} \\ &\quad + \eta \gamma (y_t - a_t) y_t - \eta(y_t - a_t) w_t - \gamma y_t w_t - \eta(y_t - a_t) q_t - \gamma y_t q_t + q_t w_t \\ &\quad - \left(\frac{1}{\theta - 1} \right) \left(\gamma \hat{\theta}_t y_t - \hat{\theta}_t q_t + \eta \hat{\theta}_t (y_t - a_t) - \hat{\theta}_t w_t \right) \\ &\quad + \hat{\theta}_t - \frac{1 + \theta}{\theta - 1} \frac{\hat{\theta}_t^2}{2} - \phi(\gamma - 1) \frac{\pi_t^{w^2}}{2} - \gamma \xi \frac{\pi_t^2}{2} \end{aligned}$$

Using the flexible price equilibrium values for real wages ($w_t^n = a_t$) and output ($y_t^n = \frac{q_t}{\gamma+\eta} + \frac{1+\eta}{\gamma+\eta} a_t$), it is straightforward to verify:

$$J_t = (\gamma + \eta) x_t - w_t^g + \frac{1}{2} ((\gamma + \eta) x_t - w_t^g)^2 + \left(\frac{1}{\theta - 1} \right) \hat{\theta}_t (w_t^g - (\gamma + \eta) x_t) \\ - \gamma \xi \frac{\pi_t^2}{2} - (\gamma - 1) \phi \frac{\pi_t^{w^2}}{2} + \hat{\theta}_t - \frac{1 + \theta}{\theta - 1} \frac{\hat{\theta}_t^2}{2}$$

So we can write the 2nd-order expansion of NKWPC as:

$$\phi(\pi_t^w) + (3\phi - \phi\psi) \frac{\pi_t^{w^2}}{2} = \frac{1}{\theta - 1} (J_t) + \beta \phi E_t \left[\pi_{t+1}^w + (5 - \psi) \frac{\pi_{t+1}^{w^2}}{2} \right] \\ + \beta \phi E_t \left[\begin{array}{c} -\gamma y_{t+1} \pi_{t+1}^w + \gamma y_t \pi_{t+1}^w + q_{t+1} \pi_{t+1}^w \\ -q_t \pi_{t+1}^w + h_{t+1} \pi_{t+1}^w - h_t \pi_{t+1}^w - \pi_{t+1} \pi_{t+1}^w \end{array} \right] \\ + \mathcal{O}(\epsilon^3)$$

Re-writing the terms inside the expectation, we have:

$$\pi_t^w + (3 - \psi) \frac{\pi_t^{w^2}}{2} = \frac{1}{\phi(\theta - 1)} (J_t) + \beta E_t [\pi_{t+1}^w] + \beta (3 - \psi) E_t \left(\frac{\pi_{t+1}^{w^2}}{2} \right) \\ + \beta E_t \left[\left(\Delta y_{t+1}^n - \Delta w_{t+1}^n - (\gamma - 1) \Delta x_{t+1} - \left(r_{t+1}^{bn} + \pi_{t+1} - \pi_{t+1}^w \right) \right) \pi_{t+1}^w \right] \\ + \mathcal{O}(\epsilon^3)$$

Next we turn to the New Keynesian Phillips Curve. Recall its nonlinear form is:

$$\frac{1 - \Phi_t^\pi}{\varepsilon - 1} + \Pi_t \frac{\partial \Phi_t^\pi}{\partial \Pi_t} = \frac{\varepsilon(1 - \varsigma)}{\varepsilon - 1} \frac{W_t}{P_t} \frac{\Upsilon_t}{A_t} + \beta E_t \left[\frac{U_c(C_{t+1}; Q_{t+1}) Y_{t+1}}{U_c(C_t; Q_t) Y_t} \frac{\partial \Phi_{t+1}^\pi}{\partial \Pi_{t+1}} \Pi_{t+1} \right]$$

Taking a 2nd-order expansion and substituting for consumption using a 2nd-order expansion of the aggregate resource constraint we have:

$$\pi_t + \frac{3}{2} \pi_t^2 = \frac{1}{\xi(\varepsilon - 1)} \left(w_t^g + \mu_t + \frac{1}{2} (w_t^g + \mu_t)^2 - \xi \frac{\pi_t^2}{2} \right) + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right] \\ + \beta E_t [-(\gamma - 1) \Delta y_{t+1} \pi_{t+1} + \Delta q_{t+1} \pi_{t+1}] + \mathcal{O}(\epsilon^3)$$

where $w_t^g := w_t - w_t^n$ and $w_t^n := a_t$. Re-writing the terms in expectations, we have:

$$\pi_t + \frac{3}{2} \pi_t^2 = \frac{1}{\xi(\varepsilon - 1)} \left(w_t^g + \mu_t + \frac{1}{2} (w_t^g + \mu_t)^2 - \xi \frac{\pi_t^2}{2} \right) + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right] \\ + \beta E_t \left[\left(\Delta y_{t+1}^n - r_{t+1}^{bn} \right) \pi_{t+1} - (\gamma - 1) \Delta x_{t+1} \pi_{t+1} \right] + \mathcal{O}(\epsilon^3)$$

where $y_t^n := \frac{1}{\gamma + \eta} q_t + \frac{1 + \eta}{\gamma + \eta} a_t$ and $\hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}$.

Finally, recall the representative household Euler equation is:

$$1 = \beta E_t \left(\frac{U_c(C_{t+1}; Q_{t+1})}{U_c(C_t; Q_t)} \left(1 + i_t^b \right) \frac{1}{\Pi_{t+1}} \right)$$

Taking a 2nd-order expansion and reformulating in terms of gaps we have:

$$\begin{aligned}
0 = & -\gamma E_t [(x_{t+1} - x_t) (1 + r_{t+1}^g)] + E_t (r_{t+1}^g) + \frac{1}{2} E_t [(r_{t+1}^g)^2] \\
& + \frac{\gamma^2}{2} E_t [(x_{t+1} - x_t)^2] + \gamma \frac{\xi}{2} E_t [\pi_{t+1}^2 - \pi_t^2] \\
& + \gamma \frac{\phi}{2} E_t [\pi_{t+1}^{w^2} - \pi_t^{w^2}] + \mathcal{O}(\epsilon^3)
\end{aligned}$$

where $r_{t+1}^g := \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn}$ and $\hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}$.

B.2.4. DEFINING THE QC PROBLEM

The QC optimal policy approximation of the NK economy with costly price and wage adjustment is to solve:

$$\min_{\{x_t, \pi_t^w, \pi_t, w_t^g, \hat{i}_t^b\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} L_t^{QC} \right]$$

$$\begin{aligned}
L_t^{QC} := & (\gamma + \eta) \left(1 + \omega_t^{QC} \right) \frac{x_t^2}{2} + \xi \left(1 + \omega_t^{QC} \right) \frac{\pi_t^2}{2} + \xi \frac{\pi_t^3}{2} - \xi (\gamma - 1) \frac{\pi_t^2 x_t}{2} + \frac{1}{2} \phi \left((\gamma - 1) \pi_t^{w^2} x_t \right) \\
& + \frac{1}{2} \phi \left(\left(1 + \omega_t^{QC} \right) \pi_t^{w^2} - \left(1 - \frac{1}{3} \psi \right) \pi_t^{w^3} - \pi_t^{w^2} w_t^g \right) - \left((\gamma - 1)^2 - (\eta + 1)^2 \right) \frac{1}{3!} x_t^3
\end{aligned}$$

and subject to the 2nd-order expansion of the NKWPC:

$$\begin{aligned}
\pi_t^w + (3 - \psi) \frac{\pi_t^{w^2}}{2} = & \frac{J_t}{\phi(\theta - 1)} + \beta E_t [\pi_{t+1}^w] - \frac{\beta}{2} (\psi - 5) E_t [\pi_{t+1}^{w^2}] \\
& - \beta E_t [(\gamma - 1) (\Delta x_{t+1} + \Delta y_{t+1}^n) \pi_{t+1}^w] \\
& + \beta E_t [(\Delta q_{t+1} - \Delta a_{t+1}) \pi_{t+1}^w - \pi_{t+1} \pi_{t+1}^w]
\end{aligned}$$

$$\begin{aligned}
J_t := & (\gamma + \eta) x_t - w_t^g + \frac{1}{2} ((\gamma + \eta) x_t - w_t^g)^2 + \left(\frac{1}{\theta - 1} \right) \hat{\theta}_t (w_t^g - (\gamma + \eta) x_t) \\
& - \gamma \xi \frac{\pi_t^{w^2}}{2} - (\gamma - 1) \phi \frac{\pi_t^{w^2}}{2} + \hat{\theta}_t - \frac{1 + \theta \hat{\theta}_t^2}{\theta - 1} \frac{1}{2}
\end{aligned}$$

and subject to the second-order expansions of the NKPC and NKIS:

$$\begin{aligned}
\pi_t + \frac{3}{2} \pi_t^2 = & \frac{1}{\xi(\epsilon - 1)} (s_t^g) + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right] + \beta E_t \left[(\Delta y_{t+1}^n - r_{t+1}^{bn}) \pi_{t+1} - (\gamma - 1) \Delta x_{t+1} \pi_{t+1} \right] \\
E_t (r_{t+1}^g) = & E_t \left[\gamma \Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g)^2 - \frac{\gamma^2}{2} (\Delta x_{t+1})^2 - \gamma \frac{\xi}{2} \Delta \pi_{t+1}^2 - \gamma \frac{\phi}{2} \Delta \pi_{t+1}^{w^2} + \gamma \Delta x_{t+1} r_{t+1}^g \right]
\end{aligned}$$

with real marginal cost, interest-rate, wage and output gaps and wage inflation defined as:

$$\begin{aligned}
s_t^g := & w_t^g + \mu_t + \frac{1}{2} (w_t^g + \mu_t)^2 - \xi \frac{\pi_t^2}{2}, & r_{t+1}^g := & \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn} \\
w_t^g := & w_t - w_t^n, & x_t := & y_t - y_t^n, & \pi_t^w := & \Delta w_t + \pi_t
\end{aligned}$$

and the associated flexible price equilibrium is characterized by:

$$y_t^n := \frac{q_t}{\gamma + \eta} + \frac{1 + \eta}{\gamma + \eta} a_t, \quad \hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}, \quad w_t^n := a_t$$

and where the state contingent weight in the loss is defined as $\omega_t^{QC} := \left(1 - \frac{(\gamma-1)}{\gamma+\eta}\right) q_t - (\gamma-1) \frac{1+\eta}{\gamma+\eta} a_t$.

Notice this approximating problem has a 3rd-order expansion of welfare where endogenous 1st-order terms have been eliminated, and is minimized subject to 2nd-order expansions of the constraint set that define the economy's decentralized equilibrium. Thus, lying within the class of $(n, n+1)$ optimal policy approximations defined in Proposition 1 with $n = 2$, the solution to it yields a 2nd-order accurate approximation of the solution to optimal Ramsey policy in the same economy – that is a 2nd-order approximation of the optimality (FOC) conditions from maximizing (B1) subject to constraints (B2) to (B5). Applying Proposition 1, when $n = 1$, and using the next lemma establishes the LQ approximation of the same problem.

B3. LQ Approximation of the NK Economy with Costly Wage and Price Adjustment

Here we state the corresponding LQ approximation of the New Keynesian economy with costly price and wage adjustment. This serves as a useful comparison to the QC approximation just derived and for deriving the results that apply under LQ in Appendix D.

LEMMA 3: *The LQ approximation of the optimal Ramsey problem is:*

$$\min_{\{\hat{i}_t^b, x_t, \pi_t, \pi_t^w, w_t^g\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(\xi \frac{\pi_t^2}{2} + (\gamma + \eta) \frac{x_t^2}{2} + \frac{1}{2} \phi \left(\pi_t^{w^2} \right) \right) \right]$$

where the minimization is subject to the 1st-order expansion of the NKWPC:

$$\pi_t^w = \frac{1}{\phi(\theta - 1)} (J_t) + \beta E_t [\pi_{t+1}^w]$$

the marginal disutility of supplying labor:

$$J_t := (\gamma + \eta) x_t - w_t^g + \hat{\theta}_t$$

the 1st-order expansions of the NKPC and NKIS curves:

$$\begin{aligned} \pi_t &= \kappa s_t^g + \beta E_t [\pi_{t+1}] \\ E_t [r_{t+1}^g] &= E_t [\gamma \Delta x_{t+1}] \end{aligned}$$

with real marginal cost, interest-rate, wage and output gaps and wage inflation defined as:

$$\begin{aligned} s_t^g &:= w_t^g + \mu_t, & r_{t+1}^g &:= \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn} \\ w_t^g &:= w_t - w_t^n, & x_t &:= y_t - y_t^n, & \pi_t^w &:= \Delta w_t + \pi_t \end{aligned}$$

the flexible price and wage equilibrium is:

$$y_t^n := \frac{q_t}{\gamma + \eta} + \frac{1 + \eta}{\gamma + \eta} a_t, \quad \hat{r}_{t+1}^{bn} := \gamma \Delta y_{t+1}^n - \Delta q_{t+1}, \quad w_t^n := a_t$$

PROOF:

This proof follows the same steps as that used to derive the QC approximation in Proposition 3. The only difference is that one only needs to use 2nd-order expansions of household felicity and the aggregate resource constraint to eliminate endogenous 1st-order terms in the objective. This modified 2nd-order objective (in this case the modified 2nd-order loss function) can then be minimized subject to 1st-order expansions NKWPC, NKPC and NKIS to derive the minimization problem above.

APPENDIX C: A QUALITATIVE COMPARISON OF LQ AND QC IN THE SIMPLIFIED NEW
KEYNESIAN ECONOMY

Here, we focus on the LQ and QC approximations in a simpler New Keynesian (NK) economy with productivity shocks, cost-push shocks and costly price adjustment only, and with household consumption risk aversion approaching 1 (log utility from consumption). We do this to present sharp analytical distinctions between the LQ and QC approximations in the simplest version of the NK model.

C1. The LQ Approximation of the Simple NK Model

First we present the LQ approximation.

LEMMA 4: *The LQ approximation of optimal policy assuming log utility ($\gamma \rightarrow 1$), no wage adjustment costs ($\phi = 0$) and only productivity and cost-push shocks are present ($\text{var}(a_t) > 0$, $\text{var}(\mu_t) > 0$) is:*

$$(C1) \quad \mathcal{L}_{t_0}^{LQ-LU} = \min_{\{\hat{i}_t^b, x_t, \pi_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(\xi \frac{\pi_t^2}{2} + (1+\eta) \frac{x_t^2}{2} \right) \right]$$

$$(C2) \quad s.t : \pi_t = \kappa s_t^g + \beta E_t [\pi_{t+1}]$$

$$(C3) \quad E_t [r_{t+1}^g] = E_t [\Delta x_{t+1}]$$

where the real marginal cost and interest rate gaps are defined by $s_t^g := (1+\eta)x_t + \mu_t$ and $r_{t+1}^g := \hat{i}_t^b - r_{t+1}^{bn} - \pi_{t+1}$, the natural interest rate and output by $r_{t+1}^{bn} := \Delta y_{t+1}^n$ and $y_t^n := a_t$ and $\kappa := \xi^{-1}(\varepsilon - 1)^{-1}$ and where the variances of demand and wage mark-up shocks are zero ($\text{var}(q_t) = \text{var}(\hat{\theta}_t) = 0$).

PROOF:

One can derive the LQ program in this case by evaluating the LQ problem in Lemma 3 (Appendix C.3) as $\gamma \rightarrow 1$, $\phi = 0$ and $\text{var}(q_t) = \text{var}(\hat{\theta}_t) = 0$. Doing so, we obtain:

$$\min_{\{x_t, \pi_t, \hat{i}_t^b\}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left((1+\eta) \frac{x_t^2}{2} + \xi \frac{\pi_t^2}{2} \right) \right]$$

subject to the first-order approximation of the NKPC and NKIS:

$$\begin{aligned} \pi_t &= \xi^{-1}(\varepsilon - 1)^{-1} s_t^g + \beta E_t [\pi_{t+1}] \\ E_t [r_{t+1}^g] &= E_t [\Delta x_{t+1}] \end{aligned}$$

where the real marginal cost, interest-rate, wage and output gaps are defined as:

$$s_t^g := (1+\eta)x_t + \mu_t, \quad r_{t+1}^g := \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn}, \quad x_t := y_t - y_t^n$$

and the associated flexible price equilibrium is characterized by:

$$y_t^n := a_t, \quad \hat{r}_{t+1}^{bn} := \Delta y_{t+1}^n$$

C2. The QC Approximation of the Simple NK Model

Next we derive the QC approximation of optimal policy in the same economy:

PROPOSITION 2: *The QC approximation of optimal policy around an efficient steady state assuming log utility ($\gamma \rightarrow 1$), no wage adjustment costs ($\phi = 0$) and only productivity and cost-push shocks are present ($\text{var}(a_t) > 0$, $\text{var}(\mu_t) > 0$) is:*

$$(C4) \quad \mathcal{L}_{t_0}^{QC-LU} = \min_{\{\hat{i}_t^b, x_t, \pi_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left(\xi \frac{\pi_t^2}{2} + (1+\eta) \frac{x_t^2}{2} + \xi \frac{\pi_t^3}{2} + (1+\eta)^2 \frac{x_t^3}{3!} \right) \right]$$

subject to the 2nd-order expansion of the constraint set:

$$(C5) \quad \pi_t + \frac{3}{2} \pi_t^2 = \kappa s_t^g + \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right]$$

$$(C6) \quad E_t [r_{t+1}^g] = E_t \left[\Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g - \Delta x_{t+1})^2 - \frac{\xi}{2} \Delta \pi_{t+1}^2 \right]$$

where the real marginal cost and interest rate gaps are given by:

$$s_t^g := (1+\eta) x_t + \mu_t + \frac{1}{2} ((1+\eta) x_t + \mu_t)^2 - \frac{\xi}{2} \pi_t^2, \quad r_{t+1}^g := \hat{i}_t^b - r_{t+1}^{bn} - \pi_{t+1}$$

and the flexible-price equilibrium natural interest rate and output by $r_{t+1}^{bn} := \Delta y_{t+1}^n$ and $y_t^n := a_t$ and $\kappa := \xi^{-1}(\varepsilon - 1)^{-1}$ and where the variances of demand and wage mark-up shocks are zero ($\text{var}(q_t) = \text{var}(\hat{\theta}_t) = 0$). While the QC approximation includes all terms, the LQ approximation includes only quadratic terms in the objective and linear terms in the constraints.

PROOF:

This is straightforward to show by evaluating the QC approximation of optimal policy in Proposition 3 as $\gamma \rightarrow 1$, $\phi = 0$ and $\text{var}(Q_t) = \text{var}(\theta_t) = 0$. Doing so, the QC optimal problem in this case is given by:

$$\min_{\{\hat{i}_t^b, x_t, \pi_t\}_{t \geq t_0}} E_{t_0} \left[\sum_{t \geq t_0} \beta^{t-t_0} \left((1+\eta) \frac{x_t^2}{2} + \xi \frac{\pi_t^2}{2} + \xi \frac{\pi_t^3}{2} + (\eta+1)^2 \frac{1}{3!} x_t^3 \right) \right]$$

subject to the 2nd-order approximations of the NKPC and NKIS:

$$\begin{aligned}\pi_t + \frac{3}{2}\pi_t^2 &= \kappa s_t^g + \beta E_t \left[\pi_{t+1} + \frac{3}{2}\pi_{t+1}^2 \right] \\ E_t [r_{t+1}^g] &= E_t \left[\Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g)^2 - \frac{1}{2} (\Delta x_{t+1})^2 - \frac{\xi}{2} \Delta \pi_{t+1}^2 + \Delta x_{t+1} r_{t+1}^g \right]\end{aligned}$$

where the real marginal cost, interest-rate, wage and output gaps are defined as:

$$\begin{aligned}s_t^g &:= (1 + \eta) x_t + (1 + \eta)^2 \frac{x_t^2}{2} + \mu_t + (1 + \eta) x_t \mu_t + \frac{\mu_t^2}{2} - \xi \frac{\pi_t^2}{2} \\ r_{t+1}^g &:= \hat{i}_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn}, \quad x_t := y_t - y_t^n\end{aligned}$$

and the associated flexible price equilibrium is characterized by:

$$y_t^n := a_t, \quad \hat{r}_{t+1}^{bn} := \Delta y_{t+1}^n$$

Applying Lemma 4 verifies the corresponding results for the LQ approximation.

The next corollary establishes that the Divine Coincidence always holds for LQ, but not for QC.

COROLLARY 1: *When cost-push shocks can perturb the economy in the future (i.e. $\text{var}(\mu_t) > 0$), the policy $\{x_t = 0, \pi_t = 0, i_t^b = E_t[r_{t+1}^{bn}]\}_{t \geq t_0}$ is optimal under LQ in response to any realized productivity shock a_{t_0} . However, this is not an optimal policy under QC.*

PROOF:

First note that the optimal policy problem under LQ is given by Lemma (4). Forming the Lagrangian, we have:

$$\Lambda_{t_0}^{LQ} = E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} \left\{ \begin{aligned} & \frac{\xi}{2} (\pi_t^2) + \frac{1}{2} [(1 + \eta)] x_t^2 \\ & - \lambda_t \left[\xi \pi_t - (\varepsilon - 1)^{-1} ((\eta + 1) x_t + \mu_t) - \xi \beta E_t (\pi_{t+1}) \right] \\ & - \varphi_t [i_t^b - \pi_{t+1} - \hat{r}_{t+1}^{bn} - \Delta x_{t+1}] \end{aligned} \right\}$$

Taking FOC and combining them with the definition of the flexible-price equilibrium:

$$(C7) \quad 0 = (1 + \eta) x_t + (\varepsilon - 1)^{-1} \lambda_t (\eta + 1)$$

$$(C8) \quad 0 = \xi \pi_t - \xi \lambda_t - \lambda_{t-1} \xi (-1)$$

$$(C9) \quad 0 = \xi \pi_t - (\varepsilon - 1)^{-1} ((\eta + 1) x_t + \mu_t) - \xi \beta E_t (\pi_{t+1})$$

and:

$$(C10) \quad 0 = i_t^b - E_t \left[\pi_{t+1} - \hat{r}_{t+1}^{bn} - \Delta x_{t+1} \right]$$

$$(C11) \quad 0 = y_t - y_t^n - x_t$$

$$(C12) \quad 0 = y_t^n - a_t$$

$$(C13) \quad 0 = \Delta y_t^n - \hat{r}_t^{bn}$$

Solving for the optimal paths on inflation and the output gap we have:

$$\begin{aligned} \pi_t &= -(\varepsilon - 1) \Delta x_t \\ \pi_t &= E_t \left[\sum_{j \geq 0} \beta^j \left(\xi^{-1} (\varepsilon - 1)^{-1} (\eta + 1) x_{t+j} + \mu_{t+j} \right) \right] \\ x_t &= -E_t \left[\sum_{j \geq 0} \left(i_t^b - \pi_{t+1+j} - \hat{r}_{t+1+j}^{bn} \right) \right] \end{aligned}$$

Assuming no cost-push shock is realized in period t_0 , and thus $E_t [\mu_{t+j}] = 0$ for all $j \geq 0$ and $t \geq t_0$, the solution for the optimal paths $\{x_t, \pi_t, i_t^b\}_{t \geq t_0}$ is then:

$$\begin{aligned} \pi_t &= -(\varepsilon - 1) \Delta x_t \\ \pi_t &= E_t \left[\sum_{j \geq 0} \beta^j (\eta + 1) x_{t+j} \right] \\ x_t &= -E_t \sum_{j \geq 0} \left[i_t^b - \pi_{t+1+j} - \hat{r}_{t+1+j}^{bn} \right] \end{aligned}$$

and the policy $\{i_t^b = E_t [\hat{r}_{t+1}^{bn}], x_t = 0, \pi_t = 0\}_{t \geq t_0}$ is indeed an optimal policy solution to (C7) to (C13) for a single realized productivity shock a_{t_0} . That is, the divine coincidence holds for productivity shocks.

We now show the same result *does not hold* for the QC approximation. Using Proposition 2, we form the Lagrangian:

$$\Lambda_{t_0}^{QC} = E_{t_0} \sum_{t \geq t_0} \beta^{t-t_0} \left[\begin{aligned} & \frac{\xi}{2} (\pi_t^3 + \pi_t^2) + \frac{1}{2} (1 + \eta) x_t^2 + \frac{1}{3!} (1 + \eta)^2 x_t^3 \\ & - \lambda_t \left(\xi \pi_t + \xi \frac{3}{2} \pi_t^2 - (\varepsilon - 1)^{-1} s_t^g - \xi \beta E_t [\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2] \right) \\ & - \nu_t \left[s_t^g - (\eta + 1) x_t - (1 + \eta)^2 \frac{x_t^2}{2} - \mu_t - (1 + \eta) x_t \mu_t - \frac{\mu_t^2}{2} + \frac{\xi}{2} \pi_t^2 \right] \\ & - \varphi_t \left[r_{t+1}^g - \left(\Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g)^2 - \frac{\gamma^2}{2} (\Delta x_{t+1})^2 - \frac{\xi}{2} \Delta \pi_{t+1}^2 + \Delta x_{t+1} r_{t+1}^g \right) \right] \end{aligned} \right]$$

The FOC at the optimum are:

$$\begin{aligned}
0 &= (1 + \eta) x_t - \frac{1}{2} \left(-(\eta + 1)^2 \right) x_t^2 + (\varepsilon - 1)^{-1} \lambda_t \left(\eta + 1 + (\eta + 1)^2 x_t + (1 + \eta) \mu_t \right) \\
0 &= 3\xi \frac{\pi_t^2}{2} + \xi \pi_t - \lambda_t (\xi + 3\xi \pi_t) - \lambda_{t-1} \xi (-1 - 3\pi_t) - \xi \pi_t (\varepsilon - 1)^{-1} \lambda_t \\
0 &= \xi \pi_t + \xi \frac{3}{2} \pi_t^2 - (\varepsilon - 1)^{-1} s_t^g - \xi \beta E_t \left[\pi_{t+1} + \frac{3}{2} \pi_{t+1}^2 \right] \\
0 &= s_t^g - (\eta + 1) x_t - \left((\eta + 1)^2 \right) \frac{x_t^2}{2} - \mu_t - (1 + \eta) x_t \mu_t - \frac{\mu_t^2}{2} + \frac{\xi}{2} \pi_t^2 \\
0 &= E_t [r_{t+1}^g] - E_t \left[\Delta x_{t+1} - \frac{1}{2} (r_{t+1}^g)^2 - \frac{1}{2} (\Delta x_{t+1})^2 - \frac{\xi}{2} \Delta \pi_{t+1}^2 + \Delta x_{t+1} r_{t+1}^g \right]
\end{aligned}$$

where the expected real interest rate gap and the flexible-price equilibrium are given by:

$$\begin{aligned}
0 &= i_t^b - E_t \left[\pi_{t+1} - \hat{r}_{t+1}^{bn} - \Delta x_{t+1} \right] - r_t^g, & 0 &= \Delta y_t^n - \hat{r}_t^{bn} \\
0 &= y_t - y_t^n - x_t, & 0 &= y_t^n - a_t
\end{aligned}$$

Again, iterating on the NKPC and the NKIS along the optimal path we have:

$$\begin{aligned}
x_t &= E_t \left[\sum_{j>0} \left(\begin{aligned} &-r_{t+1+j}^g + \Delta x_{t+1+j} - \frac{1}{2} (r_{t+1+j}^g)^2 - \frac{1}{2} (\Delta x_{t+1+j})^2 \\ &-\frac{\xi}{2} \Delta \pi_{t+1+j}^2 + \Delta x_{t+1} r_{t+1+j}^g \end{aligned} \right) \right] \\
\pi_t + \frac{3}{2} \pi_t^2 &= E_t \left[\sum_{j>0} \beta^j \left(\xi^{-1} (\varepsilon - 1)^{-1} s_{t+j}^g \right) \right] \\
s_{t+j}^g &= (\eta + 1) x_{t+j} + \mu_{t+j} + \frac{1}{2} ((1 + \eta) x_{t+j} + \mu_{t+j})^2 - \frac{\xi}{2} \pi_{t+j}^2
\end{aligned}$$

Now consider the (the flexible-price equilibrium) candidate solution $\{x_t = 0, \pi_t = 0, i_t^b = E_t [\hat{r}_{t+1}^{bn}]\}_{t \geq t_0}$ in response to a productivity shock in period t_0 , again assuming $E_t [\mu_{t+j}] = 0$ for $j \geq 0$ and $t \geq t_0$, then the NKPC and NKIS imply:

$$0 = x_t, \quad 0 = \xi^{-1} (\varepsilon - 1)^{-1} E_t \left[\sum_{j \geq 1} \beta^j \left(\frac{\mu_{t+j}^2}{2} \right) \right]$$

for all $t \geq t_0$. However, since $E_t \left[\frac{\mu_{t+j}^2}{2} \right] > 0$ for all $j \geq 1$ and $t \geq t_0$ (recalling our initial assumption that $\text{var}(\mu_t) \geq 0$), it follows that $\{x_t = 0, \pi_t = 0, i_t^b = E_t [\hat{r}_{t+1}^{bn}]\}_{t \geq t_0}$ cannot be an equilibrium solution under QC. That is, the divine coincidence does not hold for productivity shocks under QC.

APPENDIX D: A QUANTITATIVE COMPARISON OF LQ AND QC

The 2nd-order approximate solution of our nonlinear DSGE model is estimated using Simulated Method of Moments (SMM) following Kim and Ruge-Murcia (2009). We use this method to estimate the standard deviations and persistence of the shocks to the economy, the curvature of the utility function, the parameters governing price and wage adjustment costs and the monetary policy rule. We define α to be a $p \times 1$ vector of structural parameters, m_t to be a $s \times 1$ vector of empirical observations of variables whose moments we seek to match and $m_t(\alpha)$ to be the synthetic counterpart of m_t derived from simulated data generated by the model. Then, the SMM estimator, $\hat{\alpha}$, is given by

$$\hat{\alpha} = \arg \min_{\alpha \in \mathcal{A}} M(\alpha)' W M(\alpha)$$

where

$$M(\alpha) := \frac{1}{T} \sum_{t=1}^T m_t - \left(\frac{1}{\lambda T} \right) \sum_{t=1}^{\lambda T} m_t(\alpha)$$

T is the sample size, λ is a positive constant and W is a $p \times p$ weighting matrix. $M(\alpha)$ is the difference between the simulated moments and their empirical counterparts, given the parameter set α . The weighting matrix is a diagonal matrix with the inverse of the variances of the simulated moments along its main diagonal. Thus moments that are simulated with greater precision, and thus the model is more certain of, are given greater weight. For the simulation of the model, innovations are drawn independently from a standard normal distribution and the number of simulated observations is five times larger than the sample size (that is, $\lambda = 5$). We use five time series using quarterly data from 1964:II to 2006:II to generate moments to match: real per-capita consumption, hours worked, price inflation, wage inflation and the nominal interest rate. The moments in $M(\alpha)$ are all five variances, ten covariances and five autocovariances of the data series.

D1. IRFs for the Remaining Shocks

In the main text we focus on the shocks where the differences between the LQ and QC solutions are quantitatively largest. Here we report the IRFs for the remaining shocks in the model (cost-push and demand). It should be noted that while the quantitative departures from zero targets for price and wage inflation and the output gap are small in response to demand shocks (reflecting the estimated parameterization), it is not zero. For higher variances of all shocks, a non-zero response to demand shocks can be observed as well. The

quantitative differences between the LQ and QC solutions for cost-push shocks are also increasing in the shock variances as well. All impulse response functions are computed in Dynare and use pruning when evaluated at the 2nd-order.³⁹

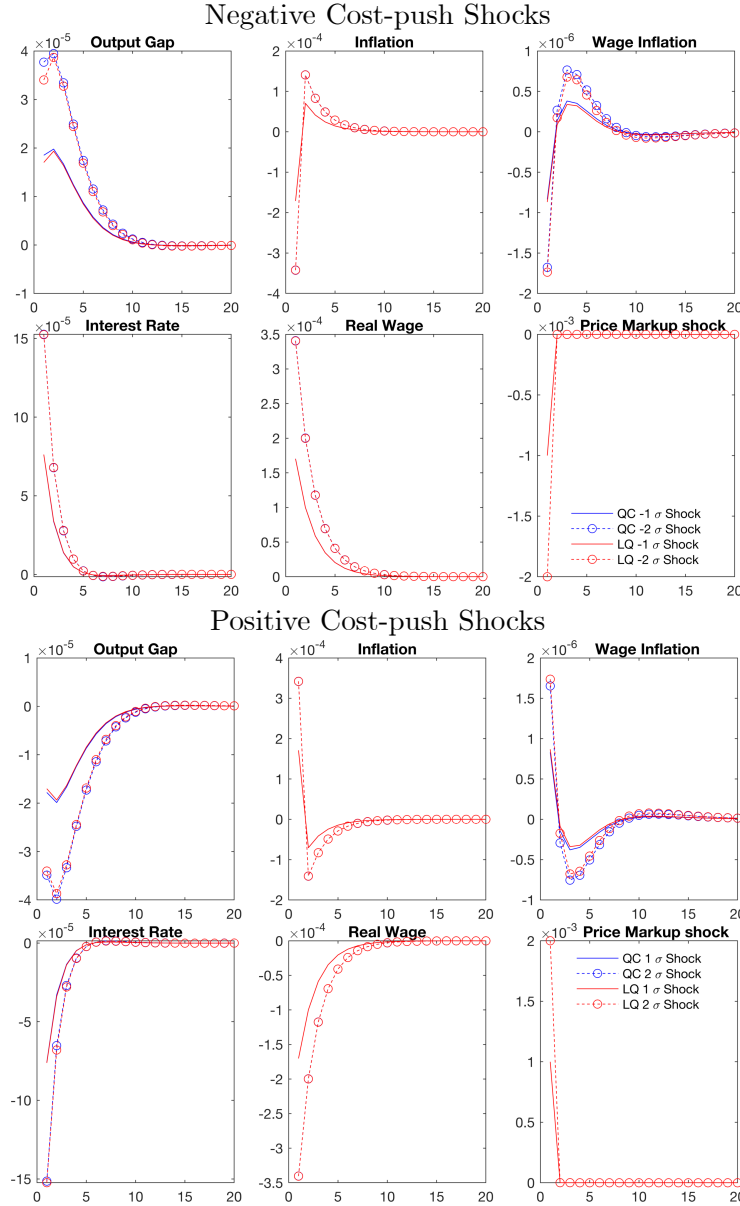


FIGURE D1. IRFS FROM THE LQ AND QC SOLUTIONS TO NEGATIVE AND POSITIVE COST-PUSH SHOCKS

Notes: 1σ and 2σ denote 1 and 2 standard deviation shocks.

³⁹See Adjemian, S., Bastani, H., Juillard, M., Karamé, F., Mailh, J., Mihoubi, Ferhat, Perendia, G., Pfeifer, J., Ratto, M. and Villemot, S. (2011), “Dynare: Reference Manual, Version 4,” Dynare Working Papers, 1, CEPREMAP.

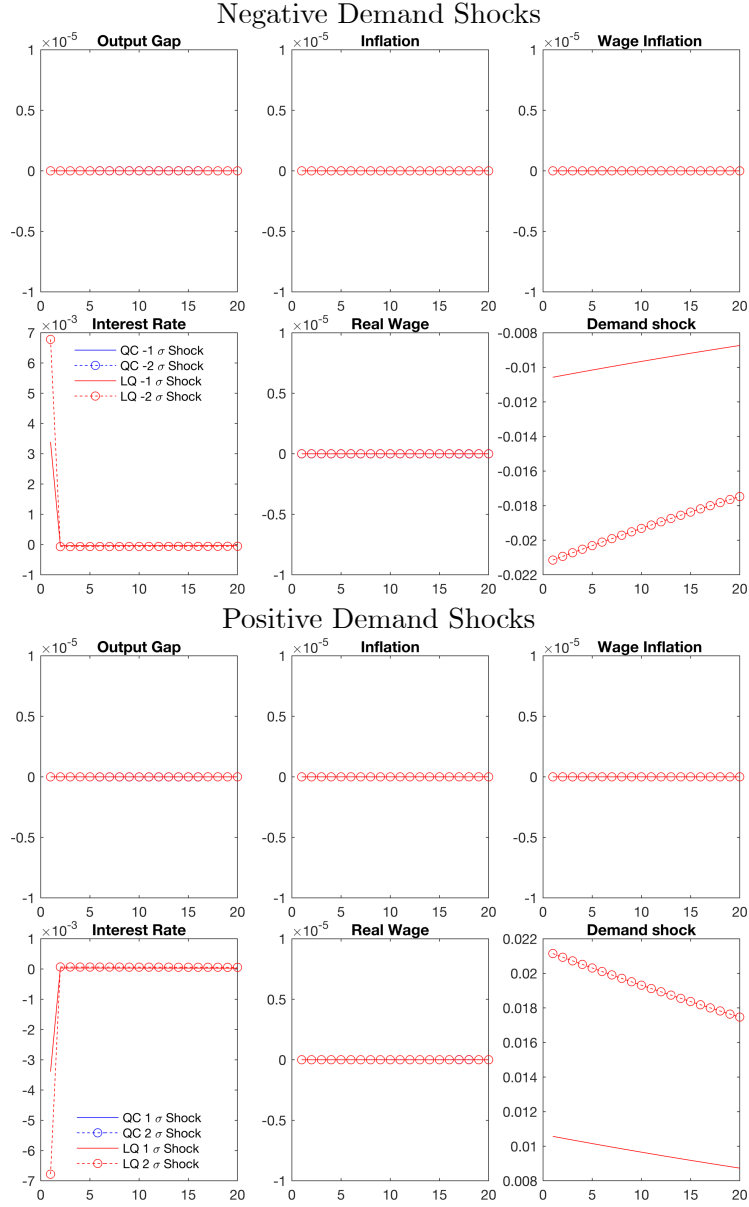


FIGURE D2. IRFs FROM THE LQ AND QC SOLUTIONS TO NEGATIVE AND POSITIVE DEMAND SHOCKS

Notes: 1σ and 2σ denote 1 and 2 standard deviation shocks.

D2. Historical Policy Counterfactuals

We use a standard particle filter as outlined by Andreasen (2011) to estimate filtered exogenous shocks over the period 1964:II to 2006:II. We assume Federal Reserve policy can be summarized by a Taylor rule and all parameters are set at their Table 1 values. We use the same empirical data ($\mathbf{g}_t$) on price inflation, wage inflation, real consumption per capita and hours worked per capita.⁴⁰ All data are seasonally adjusted at the source and are linearly detrended prior to filtering.

The standard particle filter estimates the joint posterior density of the shocks by repeatedly sampling from them and weighting the filtered predictions for observed endogenous variables according to their distance from their empirical counterpart. The latter are assumed to be measured with noise, ν_t , but otherwise equal to their model equivalents ($\tilde{\mathbf{g}}_t$). That is, $\mathbf{g}_t = \tilde{\mathbf{g}}_t + \nu_t$, where $\nu_t \sim N(0, 10^{-5})$. The particle filter thus estimates the joint historical posterior density of demand, productivity, cost-push and wage-markup shocks in each period, with the posterior mean time series for each shock shown in Figure E3.

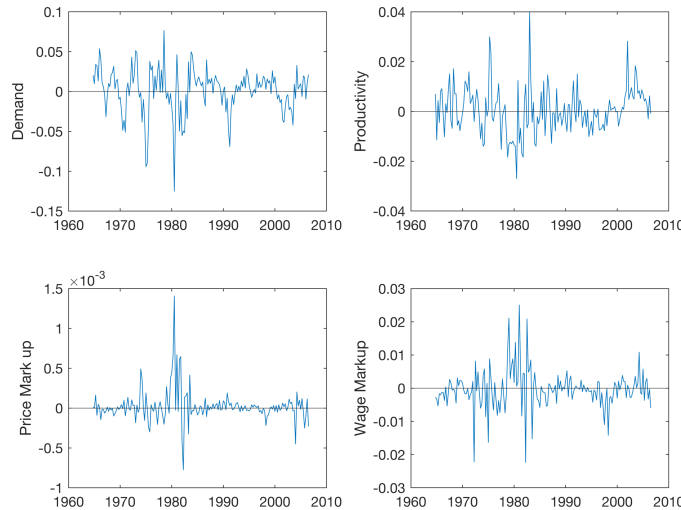


FIGURE D3. POSTERIOR MEAN HISTORICAL SHOCK ESTIMATES

Notes: The figure above reports filtered estimates for each shock evaluated at the joint posterior mean.

We then use the posterior mean estimates to simulate the historical counterfactuals under each policy regime – the empirical Taylor rule, optimal LQ policy and optimal QC policy.

⁴⁰Price and wage inflation are measured by the percentage change in the consumer price index (CPI) and the average hourly earnings for private industries, respectively. Hours worked is measured by the aggregate weekly hours index for total private industries. Real consumption is measured by personal consumption expenditure per capita deflated by the CPI.